

Discovering Ordinary Differential Equations with LLM-Based Qualitative and Quantitative Evaluation

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Abstract

Discovering governing differential equations from observational data is a fundamental challenge in scientific machine learning. Existing symbolic regression approaches rely primarily on quantitative metrics; however, real-world differential equation modeling also requires incorporating domain knowledge to ensure physical plausibility. To address this gap, we propose DoLQ, a method for discovering ordinary differential equations with LLM-based qualitative and quantitative evaluation. DoLQ employs a multi-agent architecture: a Sampler Agent proposes dynamic system candidates, a Parameter Optimizer refines equations for accuracy, and a Scientist Agent leverages an LLM to conduct both qualitative and quantitative evaluations and synthesize their results to iteratively guide the search. Experiments on multi-dimensional ordinary differential equation benchmarks demonstrate that DoLQ achieves superior performance compared to existing methods, not only attaining higher success rates but also more accurately recovering the correct symbolic terms of ground truth equations. Our code is available at <https://github.com/Bon99yun/DoLQ>.

1. Introduction

To understand complex dynamical systems, scientists have long pursued concise mathematical laws (Wigner, 1960). Differential equations are essential tools for describing the dynamic behavior of systems across various fields including physics, chemistry, and biology, and when the governing equations are known, it becomes possible to predict and control the future states of the system. However, in many

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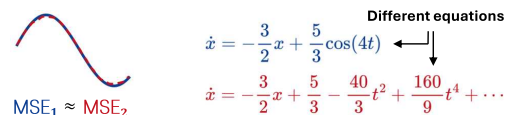


Figure 1. Similar MSE does not imply correct discovery: even with a low MSE, the identified equation may differ from the true one, suggesting that quantitative metrics alone are insufficient and that qualitative evaluation is therefore necessary.

real-world systems, the form of governing equations is not clearly known, and one must infer them from observational data. Consequently, there is a growing need for methodologies capable of deriving interpretable, explicit mathematical formulas directly from data (Rudin, 2019).

Sparse Identification of Nonlinear Dynamics (SINDy) (Brunton et al., 2016) addresses this equation discovery problem by constructing a predefined library of basis functions and identifying system dynamics via sparse regression. While efficient, SINDy struggles with selecting appropriate libraries for systems with unknown functional forms. Alternatively, Symbolic Regression (SR) approaches seek to find a symbolic function f that maps input variables x to observed outputs y , satisfying $y \approx f(x)$, from given data (Koza, 1992; Schmidt & Lipson, 2009). SR extends to dynamical system discovery by identifying the right-hand side f of an ordinary differential equation (ODE), formulated as $\dot{x} = f(t, x)$, which governs the system. SR-based methods overcome the library constraint through autonomous discovery of differential equations. For instance, ODEformer (d’Ascoli et al., 2024) leverages the Transformer architecture to generate ODEs from observational data, enabling the discovery of diverse equation structures. However, relying solely on numerical data, these methods often fail to verify the physical meaning or consistency of the generated terms.

More recently, LLMs have entered the SR landscape. These approaches replace the traditional evolutionary algorithms used for equation generation with LLMs (Shojaee et al., 2025; Grayeli et al., 2024). By leveraging the extensive prior knowledge and hypothesis generation capabilities of LLMs, these studies have outperformed traditional approaches. However, their evaluation of candidate equations still re-

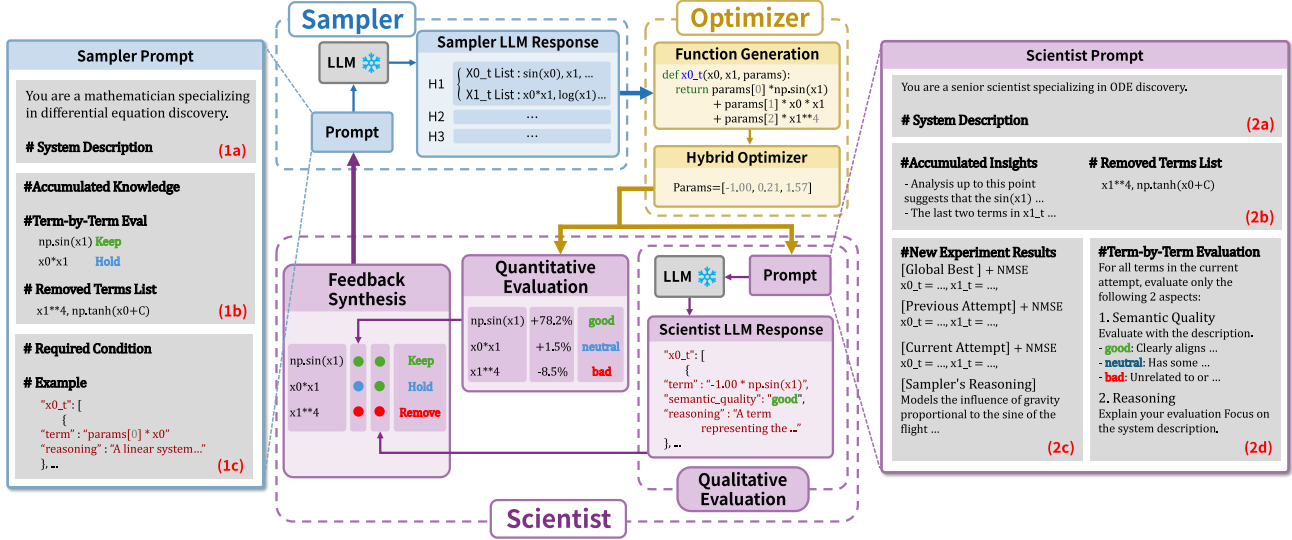


Figure 2. Overview of the DoLQ framework for LLM-based ODE discovery. The framework operates through an iterative loop among three components: (1) the Sampler Agent proposes candidate terms with physical justifications based on the system description and Scientist Agent’s feedback; (2) the Parameter Optimizer makes functions and fits their parameters; and (3) the Scientist Agent evaluates each term through qualitative semantic assessment and quantitative iterative comparison, determining actions that guide subsequent iterations.

lies primarily on quantitative criteria such as mean squared error (MSE) or model complexity. As Figure 1 illustrates, equations with similarly low numerical error can nevertheless imply different physical mechanisms. This highlights the limitation of purely numerical evaluation and motivates the need for qualitative assessment. Since differential equations in science are often grounded in natural phenomena and human intuition, the physical plausibility of discovered terms becomes essential.

To address these limitations, we propose **DoLQ**, a novel framework that **D**iscovers **O**rdinary differential equations with **L**LM-based **Q**ualitative and **Q**uantitative evaluation. DoLQ is a collaborative framework composed of three key components: a Sampler Agent, a Parameter Optimizer, and a Scientist Agent. The Sampler Agent leverages the semantic reasoning of LLMs to propose candidate terms grounded in qualitative physical justification. The Parameter Optimizer determines the optimal numerical parameters for proposed terms. Crucially, the Scientist Agent enriches the search with qualitative evaluation by assessing whether candidate equations are semantically and physically plausible rather than relying on numerical fit alone. By combining physical reasoning with numerical validation, DoLQ can identify governing laws for multi-dimensional ODEs beyond simple polynomial forms across diverse dynamical systems.

Our main contributions are summarized as follows:

- **Integrated qualitative-quantitative framework:** DoLQ explicitly embeds qualitative physical reason-

ing into the search loop, guiding the discovery process toward equations that satisfy both numerical accuracy and physical interpretability while maintaining compact equation structures.

- **Superior performance on complex systems:** We demonstrate that DoLQ achieves the highest success rates in identifying correct governing equations, particularly excelling where existing methods struggle to capture the correct dynamics.
- **Impact of qualitative evaluation:** We demonstrate the effectiveness of qualitative reasoning in filtering out physically implausible terms, validating that semantic assessment plays a crucial role in guiding the discovery process toward correct governing equations.

2. Problem Formulation

We consider a data-driven discovery task where the input consists of two parts: numerical observations and a semantic description. Let $\mathcal{D} = \{(t_i, \mathbf{x}(t_i))\}_{i=0}^{N-1}$ be the time-series state observations, where $\mathbf{x}(t_i) \in \mathbb{R}^d$ represents the system state at time t_i . Additionally, a system description $\mathcal{T}$ is provided, which represents the domain-specific prior knowledge that a practitioner would possess when approaching the discovery task.

Our goal is to discover an interpretable system of ODEs $\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x})$, where $\mathbf{f} = [f_0, \dots, f_{d-1}]^T$ is composed of functional terms proposed by an LLM that synthesizes the domain knowledge from $\mathcal{T}$ with numerical patterns in

$\mathcal{D}$. The objective is to identify $\mathbf{f}$ that minimizes the Mean Squared Error (MSE). For the j -th dimension, the integral MSE is defined as

$$\sum_{i=0}^{N-1} \left(x_j(t_i) - \left(x_j(t_0) + \int_{t_0}^{t_i} f_j(s, \mathbf{x}) ds \right) \right)^2. \quad (1)$$

Here, the integral represents the state trajectory obtained via numerical integration of the ODE system starting from the initial condition $\mathbf{x}(t_0)$.

Specifically, $\mathcal{T}$ comprises facts known before data collection and without reference to mathematical formulas, such as physical principles and qualitative assessments. While providing this context, $\mathcal{T}$ strictly prohibits specifying mathematical structures; for example, describing “air resistance” is permitted, whereas explicit terms like $-cx^2$ or $\sin(x)$ are not. This ensures the model leverages physical intuition without knowledge of the ground-truth symbolic forms.

3. Methodology

With the problem formulation in place, we now describe the DoLQ framework and its components in detail. DoLQ is an iterative framework consisting of a Sampler Agent for generating symbolic candidates, a Parameter Optimizer for estimating coefficients, and a Scientist Agent for qualitative and quantitative evaluation (Figure 2). This cycle progressively refines the discovered equations by leveraging both domain knowledge and numerical patterns, guiding the search towards physically consistent and accurate models.

3.1. Sampler Agent

The Sampler Agent is an LLM-based component that proposes candidate terms for each dimension of the system of ODEs, along with natural language justifications grounded in the system description and feedback from the Scientist Agent. See Appendix K.1 for detailed prompt examples.

Sampler prompt. The input prompt consists of three components: (1a) the role specification stating the SR task and the system description $\mathcal{T}$ providing domain-specific context and physical principles; (1b) Scientist Agent guidance, including accumulated knowledge synthesized from previous iterations, term-by-term evaluation results (*keep* or *hold*), and the set of removed term skeletons marked as ineffective; (1c) technical constraints including output format requirements and illustrative examples.

Sampler LLM response. The Sampler Agent generates ODE candidates in a structured format, pairing each symbolic term with a natural language justification grounded in the system description. Crucially, each term is formatted as an executable Python snippet utilizing state variables $\mathbf{x}$ and parameter placeholders `params[...]` (e.g.,

`params[0] * x`). As illustrated in Figure 2, the output is organized into multiple candidate hypotheses. For each hypothesis, a distinct set of terms is generated for every dimension of the system. Each term is accompanied by its physical justification and reasoning, allowing the framework to explore diverse structural assumptions while maintaining the ability to evaluate and ablate distinct physical mechanisms.

3.2. Parameter optimizer

The Parameter Optimizer takes term lists proposed by the Sampler Agent and refines them into accurate differential equations through two steps: constructing executable functions and optimizing their parameters.

Function construction. For each set of terms, we construct an executable skeleton function by instantiating the symbolic terms with trainable parameters. This function represents a candidate differential equation $f_j(t, \mathbf{x}; \boldsymbol{\theta})$ for the j -th dimension, where $\boldsymbol{\theta}$ denotes the parameters to be optimized. See Appendix B.4 for detailed examples.

Hybrid optimization. We find the optimal parameters $\boldsymbol{\theta}$ that minimize the residual MSE:

$$\sum_{i=0}^{N-1} (\dot{x}_j(t_i) - f_j(t_i, \mathbf{x}; \boldsymbol{\theta}))^2, \quad (2)$$

where $\dot{x}_j(t_i)$ denotes the numerically estimated derivative at time t_i , computed via finite differences from $\mathcal{D}$.

We adopt residual MSE instead of integral MSE (Eq. 1) for two reasons: (1) integral MSE requires costly numerical integration at each optimization step, and (2) in multi-dimensional systems, errors in one dimension cause trajectory divergence during integration, resulting in poor scores even for correctly identified dimensions. Residual MSE evaluates each dimension independently, enabling proper credit assignment. We therefore use residual MSE throughout optimization.

While many SR studies utilize BFGS (Fletcher, 2013), it is known to be sensitive to initialization and often fails to converge if the starting values are not well-posed (see Section 5.2). Therefore, we employ a hybrid strategy that first applies differential evolution (Storn & Price, 1997) to identify a promising parameter region, followed by BFGS optimization for refinement. In practice, we evaluate all three strategies—BFGS alone, differential evolution alone, and the hybrid method—selecting the candidate with the lowest MSE.

3.3. Scientist Agent

The Scientist Agent re-evaluates the optimized candidate equations $\mathbf{f}$ received from the Parameter Optimizer. It as-

sesses each candidate term from two complementary perspectives: qualitative evaluation based on the system description and quantitative contribution measured through iterative comparison. Upon completing these evaluations, the agent synthesizes both results to provide feedback and recommended actions to the Sampler Agent.

3.3.1. QUANTITATIVE EVALUATION

For quantitative evaluation, the Scientist Agent assesses which terms in the optimized equation are truly necessary. We perform a simple ablation test. For each term in the equation, we temporarily remove it by setting its coefficient to zero and recalculate the residual MSE. If the error significantly increases without the term, it is contributing meaningfully to the fit. If the error stays the same or decreases, the term is unnecessary or harmful and should be removed.

Based on this ablation analysis, terms are classified into three categories: *good* (terms whose removal significantly increases error, indicating positive contribution), *neutral* (terms whose removal has negligible impact on error), and *bad* (terms whose removal decreases error, indicating negative impact). To make the ablation criterion concrete, consider the illustrative example shown in the middle panel of Figure 2. In that example, $\text{np.sin}(x_1)$ is classified as *good* because removing it increases the MSE by 78.2%, whereas x_1**4 is classified as *bad* because its removal decreases the overall error, indicating that the term was inflating the fit. See Appendix B for details.

3.3.2. QUALITATIVE EVALUATION

For qualitative evaluation, the Scientist LLM assesses how well each term aligns with the physical or mathematical meaning described in the system description. See Appendix K.2 for details.

Scientist prompt. The input prompt to the Scientist Agent comprises four components: (2a) context information including the current iteration progress and system description, enabling the agent to understand the search trajectory and physical background; (2b) accumulated insights from previous evaluations and constraints listing previously removed term structures to prevent redundant proposals in subsequent iterations; (2c) experimental results presenting the Global Best equation, Previous Attempt, and Current Attempt, each with per-dimension MSE values, optimized coefficients, and the Sampler’s physical justifications for each proposed term, allowing the agent to assess performance improvements and evaluate the physical plausibility of new terms relative to established baselines; (2d) evaluation instructions specifying semantic quality grading criteria and requirements for physical reasoning, ensuring consistent evaluation standards across iterations.

Table 1. Benchmark ODEs and their temporal domains for ID and ID-Ext evaluation.

Benchmark	Equation	Time Domain	
		ID	ID-Ext
SIR(2D)	$\dot{x}_0 = -0.4x_0x_1$ $\dot{x}_1 = 0.4x_0x_1 - 0.314x_1$	[0, 2]	[0, 4]
CDIMA(2D)	$\dot{x}_0 = 8.9 - \frac{4.0x_0x_1}{x_0^2+1} - x_0$ $\dot{x}_1 = 1.4x_0 \left(1 - \frac{x_1}{x_0^2+1}\right)$	[0, 5]	[0, 10]
Glider(2D)	$\dot{x}_0 = -x_0^2/5.0 - \sin(x_1)$ $\dot{x}_1 = x_0 - \cos(x_1)/x_0$	[0, 5]	[0, 10]
Glider(4D)	$\dot{x}_0 = -9.81 \sin(x_1) - 0.030625x_0^2$ $\dot{x}_1 = -\frac{9.81 \cos(x_1)}{x_0} + 0.6125x_0$ $\dot{x}_2 = x_0 \cos(x_1)$ $\dot{x}_3 = x_0 \sin(x_1)$	[0, 5]	[0, 10]

Scientist LLM response. The Scientist Agent produces a structured response containing two components. The first component, term evaluations, provides dimension-specific assessments where each term receives a semantic quality grade paired with physical reasoning. Terms are assigned *good* if they clearly align with physical principles described in the system description, *neutral* if they are partially relevant but lack a clear physical basis, or *bad* if they lack physical meaning or contradict the system description. The second component, updated insight, synthesizes the current analysis into accumulated knowledge, identifying captured physical mechanisms and aspects requiring further investigation. This structured format enables systematic tracking of modeling progress.

3.3.3. FEEDBACK SYNTHESIS.

The final action for each term is determined by combining both evaluations: (1) if semantic quality is *bad*, immediately *remove* regardless of performance impact; (2) if both semantic quality and performance impact are *good*, assign *keep*; (3) all other combinations are assigned *hold*. Terms receiving *hold* for two consecutive iterations are converted to *remove*. Removed term skeletons, where parameters are replaced with placeholder symbols, are recorded to prevent re-proposal. To avoid permanently discarding potentially valid terms, a probabilistic forgetting mechanism periodically clears these entries, allowing re-exploration of previously rejected candidates. The detailed algorithm and decision logic are provided in Appendix B.3.

4. Experiments

4.1. Benchmark and dataset

To evaluate our method, we select seven ODE problems from ODEbench (d’Ascoli et al., 2024), a comprehensive benchmark for ODE discovery, with their physical descriptions and domain knowledge supplemented by referencing

Table 2. Quantitative performance comparison measured by NMSE. We report the dimension-averaged results. Bold with underline indicates the best, bold indicates the second best. NaN indicates solver failure or numerical instability.

Model	SIR(2D)				CDIMA(2D)				Glider(4D)			
	Residual		Integral		Residual		Integral		Residual		Integral	
	ID	ID-Ext	ID	ID-Ext	ID	ID-Ext	ID	ID-Ext	ID	ID-Ext	ID	ID-Ext
ICSR	2.80e-8	4.62e-9	1.42e-8	1.19e-8	5.16e-4	6.06e-5	3.78e-4	2.56e-4	2.55e-2	1.16e0	2.55e-1	3.19e0
LASR	1.78e-8	7.19e-9	7.60e-9	8.61e-9	2.91e-1	1.14e1	2.90e-1	4.85e1	2.50e-2	3.52e0	4.94e-1	1.00e3
LLM-SR	1.65e-8	4.14e-9	1.61e-8	2.79e-8	2.67e-3	5.08e-3	3.14e-3	9.38e-3	9.95e-7	1.12e-6	8.54e-7	4.76e-6
EDL	1.86e1	NaN	1.40e1	NaN	1.07e5	NaN	1.33e5	NaN	1.48e4	NaN	1.37e4	NaN
DoLQ(ours)	1.71e-8	3.98e-9	1.55e-8	2.63e-8	2.29e-8	2.69e-8	2.35e-8	1.15e-7	1.22e-6	1.53e-6	8.29e-7	5.21e-6

Wikipedia and related literature. For four-dimensional systems, we additionally construct benchmark problems following the same approach. Thus, we use a total of eight ODE datasets. We focus on four representative systems for detailed analysis in the main experiments. We use three systems for quantitative performance comparison as shown in Table 2: SIR(2D), CDIMA(2D), and Glider(4D). Additionally, we use Glider(2D) for structural equation analysis as shown in Figure 3. Details on the complete benchmark suite are provided in Appendix A.

Benchmark problems. Table 1 shows four representative systems from the benchmark suite. We categorize problems by their functional complexity: systems with polynomial terms versus those with rational, trigonometric, or exponential terms. Table 2 presents quantitative evaluation on three systems: SIR(2D) models disease spread, CDIMA(2D) models chemical kinetics with nonlinear saturation terms, and Glider(4D) captures flight dynamics. Figure 3 shows structural equation comparison on a dimensionless Glider(2D) variant for clearer visualization.

Extended in-domain evaluation. To assess the generalization capability of discovered equations, we build upon the evaluation framework of Shojaaee et al. (2025), which proposes in-domain (ID) and out-of-domain (OOD) test sets. While OOD evaluation is ideal for testing generalization, it is impractical in our setting because integral MSE-based validation requires initial conditions that may be unavailable in the extended regime. To address this limitation, we introduce extended-ID (ID-Ext) test sets that extend the temporal or state-space coverage while retaining observed initial conditions, enabling rigorous testing of whether discovered equations maintain predictive accuracy over longer horizons.

Data generation. State trajectories for both ID and ID-Ext regimes were generated using the fourth-order Runge–Kutta method. The time derivatives $\dot{x}$ in Eq. (2) were obtained through numerical differentiation based on the finite difference method. For each ODE, we prepared textual descriptions by referencing relevant research literature, with equation names and explicit formula structures removed

to prevent direct inference of the symbolic form. The ID regime is used for model training and selection, while the ID-Ext regime serves to evaluate extrapolation capability under extended temporal or spatial ranges. The time domain ranges for ID and ID-Ext in Table 1 were selected based on the temporal intervals where the data distribution exhibited the most significant differences, ensuring rigorous evaluation of extrapolation capability.

4.2. Baseline models and experimental setup

We evaluate DoLQ against representative LLM-based SR methods: ICSR (Merler et al., 2024), LASR (Grayeli et al., 2024), LLM-SR (Shojaaee et al., 2025), and EDL (Du et al., 2024). We selected these baselines based on their capability to incorporate textual descriptions of the system $\mathcal{T}$ via prompts and their ability to generate symbolic ODE expressions. Notably, among these baselines, LLM-SR and EDL were originally proposed and evaluated specifically for symbolic discovery of ordinary differential equations. The remaining methods, LASR and ICSR, were primarily designed for general symbolic regression tasks; nevertheless, we adapt them to our problem setting to enable a fair comparison on ODE discovery.

Most existing SR methods target scalar-valued functions ($f : \mathbb{R}^d \rightarrow \mathbb{R}$), whereas our task requires discovering vector-valued functions ($f : \mathbb{R}^d \rightarrow \mathbb{R}^d$) for ODE systems. To ensure fair comparison, we adapted the baselines to support vector-valued outputs by modifying their prompts and providing the same system description $\mathcal{T}$. All experiments were conducted using Gemini 2.5 Flash Lite (Team et al., 2023). To ensure equivalent LLM call conditions for a fair comparison, we configured the search budget as follows. For DoLQ, we perform 100 iterations. In contrast, baseline models LLM-SR, EDL, and ICSR are configured to perform 100 iterations per dimension, with LASR set to a comparable expected number of iterations per dimension. A similar budget strategy was maintained for 4D systems; further details are provided in Appendix B.

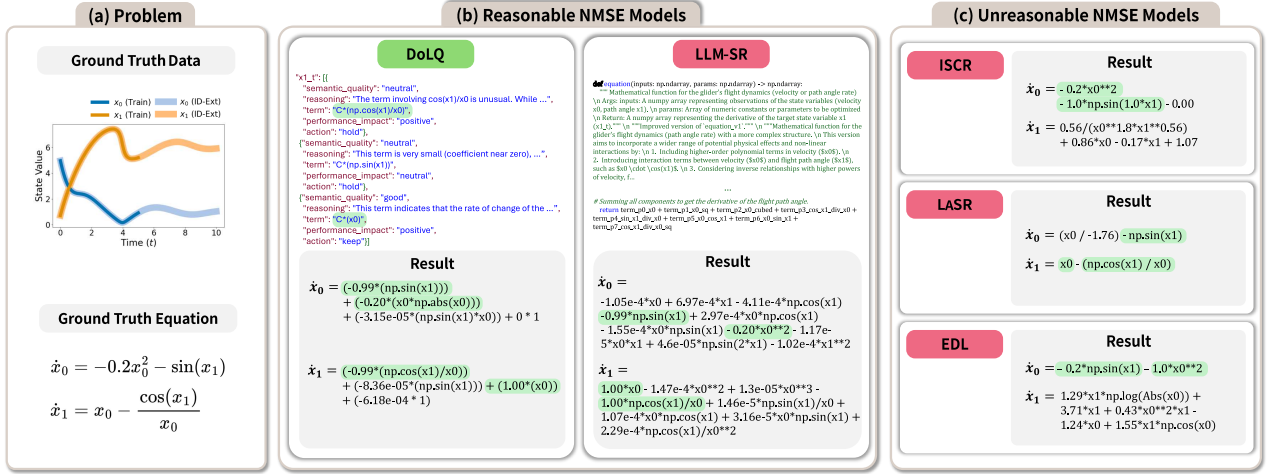


Figure 3. Equation comparison on the 2D dimensionless Glider system. Ground truth terms are highlighted in green. Group (b) shows models with reasonable NMSE: DoLQ achieves integral NMSE $< 10^{-3}$ across all dimensions, and LLM-SR exhibits the lowest NMSE among the remaining methods. Group (c) shows models with higher NMSE. Each model’s discovered equation terms and their corresponding optimized coefficients are displayed.

4.3. Performance comparison

We use normalized mean squared error (NMSE) to enable fair comparison across different problems with varying scales. We evaluate performance using two metrics: integral NMSE and residual NMSE, with the results summarized in Table 2. For the SIR(2D) system, since the equation form is readily inferable from the description, most models achieve low error rates, successfully identifying the governing dynamics. Notably, many models identify the correct mathematical structure practically from the first iteration, demonstrating that existing methods are capable when the problem structure is relatively straightforward.

However, the CDIMA(2D) system presents a distinct challenge due to its complex nonlinear saturation terms. In this case, a significant divergence emerges between residual NMSE and integral NMSE. Baseline models often exhibit low residual NMSE yet high integral NMSE, indicating a failure to capture the true underlying physics. Only DoLQ consistently achieves superior performance with stable integral NMSE scores, demonstrating its effectiveness in discovering equations with complex functional forms.

For the Glider(4D) system, both LLM-SR and DoLQ successfully identify the correct governing equations with low NMSE.

4.4. Structural comparison

Beyond the three representative systems in Table 1, we further analyze the learning outcomes on an additional ODE problem: a 2D dimensionless variant of the Glider system. This simplified version retains the essential nonlinear trigonometric dynamics while enabling clearer visualization

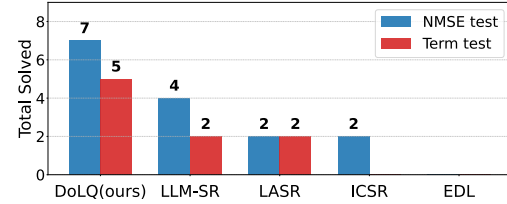


Figure 4. Success scores measured across eight benchmark ODEs. NMSE test: success is achieved if the integral NMSE across all dimensions is $< 10^{-3}$. Term test: success is achieved if the discovered equation matches the ground truth structure after excluding terms with negligible impact.

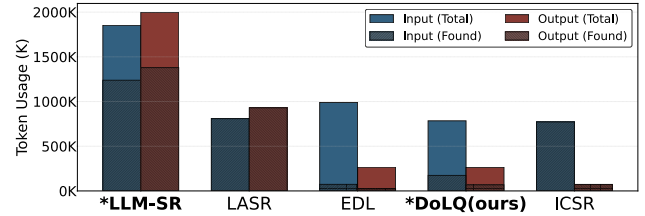


Figure 5. Cumulative token usage on the 4D Glider system. Stars (*) indicate that LLM-SR and DoLQ successfully identified the governing equation. NMSE values reach their minimum on the Glider system.

and interpretation of the discovered terms compared to the full 4D system.

Qualitative examination reveals critical differences in the discovered physical laws. In Figure 3, we categorize the results on this system into groups with reasonable NMSE (b) and those without (c). Both DoLQ and LLM-SR successfully identify all ground truth terms, whereas other baselines capture only partial dynamics. However, LLM-SR often pro-

poses equation skeletons containing numerous unnecessary terms. In contrast, DoLQ proposes significantly fewer terms while successfully including all ground truth components. This efficiency stems from the Scientist Agent’s rigorous quantitative and qualitative assessment; beneficial terms receive positive evaluations, while detrimental terms are explicitly identified and removed, preventing their recurrence in subsequent iterations. Consequently, DoLQ achieves superior performance with a more focused and efficient search process.

4.5. Overall comparison and discussion

As shown in Figure 4, DoLQ achieves the highest success rate across all evaluation criteria, outperforming baseline methods on both NMSE test and Term test across diverse problem types.

Analyzing baseline characteristics reveals distinct limitations. LASR demonstrates reasonable performance on problems with simple polynomial terms, producing clean equations with minimal unnecessary components. However, the evolutionary algorithm’s search capability is fundamentally limited, failing to discover governing equations for complex systems involving non-polynomial functional forms. LLM-SR achieves moderate NMSE performance (4/8 success rate), yet shows limited success in capturing the correct equation structure. This discrepancy arises from proposing excessive candidate terms, which not only complicates parameter optimization but also generates unnecessarily verbose prompts. As evidenced by Figure 5, LLM-SR consumes substantially more tokens than DoLQ, with detailed statistics provided in Appendix C.4.

DoLQ addresses these fundamental limitations through the Scientist Agent’s evaluation framework. By rigorously assessing both qualitative evaluation and quantitative evaluation of each proposed term, the Scientist Agent effectively filters out unnecessary candidates early in the search process. This prevents the bloated equation skeletons that plague LLM-SR while maintaining the capacity to discover complex functional forms beyond the reach of evolutionary methods. The result is a framework that achieves superior accuracy across both simple and complex systems, consistently identifying the correct governing equations where baseline methods fail. DoLQ’s multi-agent architecture not only discovers physically consistent governing equations with higher success rates but also enables more interpretable results through compact equation structures. By combining qualitative reasoning with quantitative validation, DoLQ establishes a new standard for reliable and accurate ODE discovery in dynamical systems, demonstrating robust performance across diverse problem types while maintaining computational efficiency as an additional benefit.

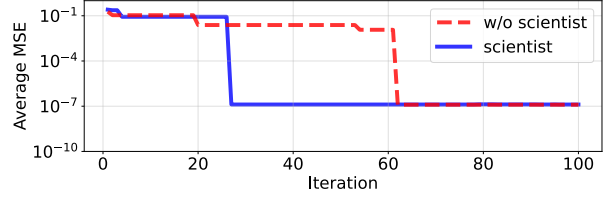


Figure 6. Impact of the Scientist Agent on search convergence for the Glider(2D) problem. DoLQ with the Scientist Agent discovers the correct equation at iteration 27, while the baseline without the Scientist Agent finds it at iteration 62, demonstrating that qualitative feedback accelerates convergence.

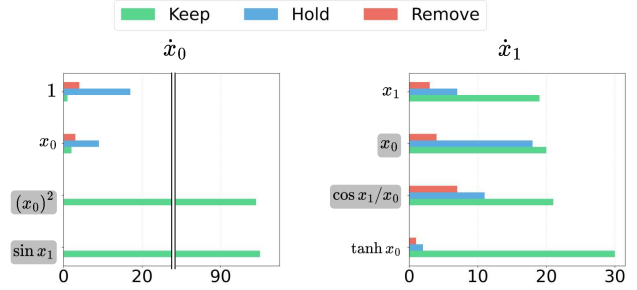


Figure 7. Action frequencies for the top-ranked terms in each target dimension during DoLQ execution on the Glider(2D) problem. Highlighted terms appear in the ground-truth equation and consistently receive high *keep* frequency, indicating that the Scientist Agent effectively preserves physically meaningful terms.

5. Component validation

5.1. Necessity of the Scientist Agent

To demonstrate the critical role of qualitative reasoning, we conduct an ablation study comparing DoLQ against a baseline variant labeled “w/o Scientist,” where the Scientist Agent is removed from the loop. In this configuration, the Sampler Agent directly receives the equation skeletons, optimized coefficients, and NMSE scores from the Parameter Optimizer as part of the prompt for the next iteration, bypassing the qualitative evaluation step. As illustrated in Figure 6, the full DoLQ framework containing the Scientist Agent exhibits significantly faster convergence than the baseline. An analysis of the reasoning logs confirms that the Scientist Agent plays a pivotal role in the discovery process; its feedback effectively filters out physically implausible terms early in the search, guiding the model toward the correct governing equations more efficiently.

Unless otherwise noted, all methods were executed three times under the same configuration, and we report the best run; Figure 6 shows one representative convergence trace generated under this protocol rather than an average over runs. Specifically, the plotted DoLQ trace corresponds to the best of these three runs.

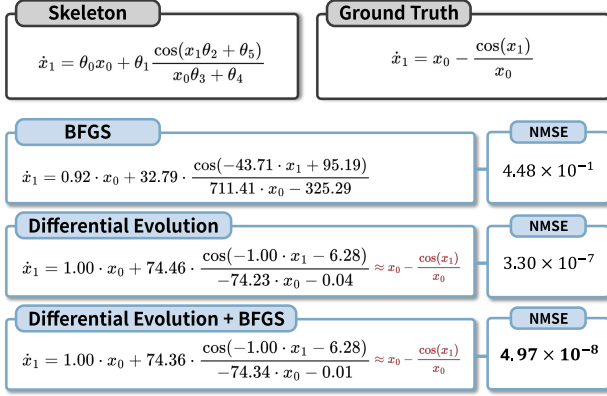


Figure 8. BFGS alone can fail even with a structurally correct skeleton, whereas the hybrid optimizer successfully recovers the correct solution. This illustrates why hybrid parameter optimization is necessary.

To further understand how the Scientist Agent operates within DoLQ, we analyze its decision patterns during the discovery process. Figure 7 reveals that ground truth terms consistently receive high *keep* frequencies, demonstrating the Scientist Agent’s effectiveness in identifying physically relevant terms. Notably, even when these terms occasionally receive *remove* recommendations, they reappear frequently across iterations due to the probabilistic sampling mechanism that considers keep/remove probabilities, global best information, and previous iteration history, enabling robust convergence toward correct equations.

5.2. Effectiveness of hybrid parameter optimization

Our choice of combining differential evolution with BFGS is motivated by the need for robust parameter estimation in the rugged loss landscapes of symbolic regression. While the LLM may propose a structurally correct equation skeleton, standard gradient-based optimization like BFGS often fails to identify the optimal parameters due to its sensitivity to initialization and susceptibility to local minima. As shown in Figure 8, even when the correct functional form is provided, BFGS yields suboptimal parameter estimation, whereas our hybrid approach—combining the global search of differential evolution with the local refinement of BFGS—successfully locates the global optimum. Crucially, the successful optimization by differential evolution verifies that the skeleton proposed by the LLM is indeed structurally similar to the ground truth. Relying solely on BFGS could lead to the incorrect rejection of a reasonable skeleton due to optimization failure, concealing the true quality of the proposed structure. We provide further analysis on the adoption frequency of each optimizer across different problem complexities in Appendix D.

6. Related work

Symbolic regression. SR aims to recover governing equations of dynamical systems in the form of explicit mathematical formulas, evolving from evolutionary algorithms (Cranmer, 2023) to incorporate physical constraints (Udrescu & Tegmark, 2020), reinforcement learning (Petersen et al., 2021; Mundhenk et al., 2021; Xu et al., 2024), and neural sequence models (Biggio et al., 2021; Kamienny et al., 2022; Valipour et al., 2021; Shojaee et al., 2023; Yu et al., 2025; Xiang et al., 2026; Huang et al., 2025). For dynamical systems, recent methods include grammar-based discovery (Omejc et al., 2024) and Transformer-based architectures (d’Ascoli et al., 2024). Recent advances include symbolic-numeric pre-training (Meidani et al., 2024) and interactive offline reinforcement learning (Tian et al., 2025). However, most existing methods prioritize point-wise or derivative-based fitting metrics without validating long-horizon trajectory consistency. While functional SR has been extensively studied with established benchmarks (Zhang et al., 2012), ODE-specific benchmarks (Matsubara et al., 2022) remain relatively limited, requiring further research (Oliveira et al., 2018).

LLMs for scientific discovery. LLMs leverage encoded scientific knowledge for equation discovery, optimizing skeletons via evolutionary search (Shojaee et al., 2025), self-improvement (Du et al., 2024), in-context learning (Merler et al., 2024), concept libraries (Grayeli et al., 2024), and symbol-numeric pre-training (Meidani et al., 2024). Recent work shows LLMs can interpret discovered expressions (Guo et al., 2025) and diagnose errors through multi-agent reasoning (Park et al., 2026), while formal verification enables structured validation (Sun et al., 2025). Beyond SR, LLMs translate natural language to formal mathematics through autoformalization and reasoning (Soroco et al., 2025) and automate research pipelines (Lu et al., 2024). However, evaluation still relies on derivative matching rather than trajectory integration, and LLMs’ semantic reasoning capacity for validation remains underutilized.

Modeling and evaluation of dynamical systems. Deep learning approaches for differential equation modeling (Czarnecki et al., 2017; Dupont et al., 2019; Greydanus et al., 2019; Raissi et al., 2019; Rubanova et al., 2019; Cranmer et al., 2019; Karniadakis et al., 2021; Linot et al., 2023; Lee et al., 2025) achieve robust predictions but lack interpretability. In contrast, SR discovers interpretable governing equations in closed form. Our approach integrates LLM-based physical reasoning into symbolic discovery to improve equation accuracy.

7. Conclusion

This paper presents DoLQ, a multi-agent framework integrating qualitative and quantitative evaluation for discovering governing ODEs. By employing LLMs to validate physical plausibility, DoLQ demonstrates substantial improvements in discovery success rates and equation interpretability, particularly for systems with complex functional forms. The systematic integration of semantic reasoning with numerical validation enables consistent identification of correct governing equations across diverse problem types, achieving the highest performance on comprehensive benchmarks. Several limitations warrant further investigation: parameter optimization depends on numerical differentiation, introducing sensitivity to measurement noise, and reasoning fixation occasionally causes premature convergence to implausible hypotheses. Future research will address these through integral-based parameter estimation and diversified hypothesis generation with refined feedback mechanisms. While ODEs benefit from established numerical solvers, extension to PDEs presents considerable difficulty due to the absence of universal solution methods. We plan to investigate advances in PDE solver research to enable reliable equation discovery in spatiotemporal dynamical systems.

Impact Statement

This work advances automated scientific discovery by enabling interpretable equation learning from observational data. The enhanced ability to discover governing equations can accelerate research in physics, biology, and engineering, reducing dependence on domain expertise for model development. However, practitioners should remain cautious of potential risks: automated discovery systems may generate spurious correlations or physically implausible models if deployed without proper validation. We emphasize the importance of human oversight in applying discovered equations to real-world systems, particularly in safety-critical domains.

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A. Dataset details

A.1. Benchmark problems and ground truth equations

Table 3 provides the complete specifications for all eight benchmark ODE systems used in our experiments. Problems ID 1-7 are sourced from ODEbench (d’Ascoli et al., 2024), while ID 8 is a 4D variant of the Glider problem, constructed to evaluate performance on four-dimensional systems with non-polynomial terms. For each benchmark problem, we list its identifier (ID 1-8), descriptive title, ground truth governing equations in mathematical form, and the temporal domains used for both ID and ID-Ext evaluation regimes. These specifications serve as the reference for comparing the equations discovered by DoLQ and baseline methods, as detailed in Table 11.

Table 3. Benchmark ODEs and their ground truth governing equations and temporal domains used for in-domain (ID) and extended in-domain (ID-Ext) evaluations.

ID	Title	Ground Truth Equations	t (ID)	t (ID-Ext)
1	SIR infection model only for healthy and sick	$\dot{x}_0 = -0.4x_0x_1$ $\dot{x}_1 = 0.4x_0x_1 - 0.314x_1$	[0, 2]	[0, 4]
2	Glider (dimensionless)	$\dot{x}_0 = -x_0^2/5.0 - \sin(x_1)$ $\dot{x}_1 = x_0 - \cos(x_1)/x_0$	[0, 5]	[0, 10]
3	Reduced model for chlorine dioxide-iodine-malonic acid reaction (dimensionless)	$\dot{x}_0 = 8.9 - \frac{4.0x_0x_1}{x_0^2+1.0} - x_0$ $\dot{x}_1 = 1.4x_0 \left(1.0 - \frac{x_1}{x_0^2+1.0}\right)$	[0, 5]	[0, 10]
4	Isothermal autocatalytic reaction model by Gray and Scott 1985 (dimensionless)	$\dot{x}_0 = 0.5(1.0 - x_0) - x_0x_1^2$ $\dot{x}_1 = -0.02x_1 + x_0x_1^2$	[0, 2]	[0, 4]
5	Interacting bar magnets	$\dot{x}_0 = 0.33 \sin(x_0 - x_1) - \sin(x_0)$ $\dot{x}_1 = -0.33 \sin(x_0 - x_1) - \sin(x_1)$	[0, 2]	[0, 4]
6	Binocular rivalry model (no oscillations)	$\dot{x}_0 = -x_0 + (e^{4.89x_1-1.4} + 1.0)^{-1}$ $\dot{x}_1 = -x_1 + (e^{4.89x_0-1.4} + 1.0)^{-1}$	[0, 2]	[0, 4]
7	Oscillator death model by Ermentrout and Kopell (1990)	$\dot{x}_0 = 1.432 + \sin(x_1) \cos(x_0)$ $\dot{x}_1 = 0.972 + \sin(x_1) \cos(x_0)$	[0, 4]	[0, 8]
8	Glider (physical units)	$\dot{x}_0 = -9.81 \sin(x_1) - 0.030625x_0^2$ $\dot{x}_1 = -\frac{9.81 \cos(x_1)}{x_0} + 0.6125x_0$ $\dot{x}_2 = x_0 \cos(x_1)$ $\dot{x}_3 = x_0 \sin(x_1)$	[0, 5]	[0, 10]

A.2. Natural language descriptions for benchmark problems

Table 4 provides the natural language descriptions ($\mathcal{T}$) for each benchmark ODE system. These descriptions contain domain-specific context and physical principles that are provided as input to the LLM-based agents during the equation discovery process.

Table 4. Natural language descriptions for benchmark problems.

ID	Natural Language Description ($\mathcal{T}$)
1	This model describes disease spread by dividing the population into susceptible and infected groups, with transmission proportional to their contact and infected individuals recovering at a constant rate. An epidemic occurs only when the transmission rate exceeds a threshold, revealing a critical condition for outbreak or extinction
2	The glider is viewed as an idealized system whose motion is expected to arise from the interplay of gravity and aerodynamic forces, with speed and flight path angle evolving from assumed initial conditions
3	One state variable represents the activator species concentration, which autocatalytically accelerates the reaction and has limited spatial mobility due to polymer indicator binding. Another represents the inhibitor species concentration that suppresses activator production. The activator production rate shows saturation beyond a certain level due to substrate limitations. The consumption process is negligible at low activator concentrations but changes sharply above a specific range. External halide addition directly reacts with the inhibitor, reducing its amount.
4	This describes an open chemical system where a precursor is continuously supplied and an autocatalytic species both self-amplifies and decays. The interplay of local activation and global depletion leads to spontaneous pattern formation such as spots, waves, spirals, and chaotic spatial behavior arising from instability of an initially uniform state
5	This system describes two nearby bar magnets whose orientations influence each other through distance-dependent magnetic interactions. Depending on initial conditions and external effects, the magnets may settle into stable alignments, oscillate, rotate together, or follow complex motion patterns
6	This describes binocular rivalry, where two competing neural populations represent different images seen by each eye and suppress each other. Depending on initial conditions and input strength, one population becomes dominant, leading to stable perception of only one image at a time
7	This describes a counterintuitive effect where coupling two identical oscillating systems can completely suppress their oscillations. Strong symmetric interaction forces both systems into a shared steady state, causing oscillatory activity to disappear regardless of initial motion
8	This is conducted to measure the physical motion characteristics of a glider in actual flight. The experimenter sets the glider’s travel speed, flight path angle relative to the horizontal plane, horizontal distance, and altitude as main measurement variables. Specifically, the state variables are defined as forward velocity, flight path angle, horizontal position, and vertical altitude. The experiment is based on an aircraft of specific mass moving through the atmosphere under constant gravitational acceleration, experiencing lift and drag determined by wing configuration and air density. The aircraft’s mass, wing area, and atmospheric conditions are defined beforehand, and the flight trajectory is recorded by setting initial launch velocity and altitude.

To reduce equation-level leakage, we removed equation names, explicit symbolic forms, and direct function-form clues from all descriptions before prompting. This design does not prove the absence of memorization in a strict causal sense, but it prevents the prompt from containing the answer in symbolic form and forces discovery to depend on consistency between the description, numerical optimization, and iterative term evaluation.

A.3. Data generation procedure

We generate synthetic trajectory data for all benchmark systems using numerical integration. The data generation process consists of three main stages: ODE solving, derivative computation, and dataset partitioning.

Numerical integration. State trajectories are generated using the Runge–Kutta method. For each ODE system specified in Table 3, we integrate the ground truth equations over the temporal domains indicated in the table.

Derivative computation. We compute time derivatives $\dot{x}(t)$ using two complementary methods to validate numerical consistency. First, we evaluate the derivatives directly by substituting the interpolated state values into the ground truth ODE functions: $\dot{x}(t_i) = f(t_i, x(t_i))$, where f represents the right-hand side of the ODE system. This provides the analytically correct derivative values denoted as $\dot{x}_{\text{gt}}$. Second, we compute numerical derivatives using NumPy’s gradient function with second-order accurate edge handling (`edge_order=2`), which applies the central difference scheme $\dot{x}(t_i) \approx (x(t_{i+1}) -$

$x(t_{i-1}))/ (2\Delta t)$ for interior points and forward/backward differences at boundaries. These numerically differentiated values, denoted as $\dot{x}_t$, serve as the target derivative values for residual-based optimization (Eq. 2 in the main text). For evaluation, we measure Residual NMSE by comparing discovered equations’ predictions against $\dot{x}_{\text{gt}}$, ensuring consistency with the ground truth dynamics.

Dataset partitioning. Generated trajectories are partitioned into two distinct temporal regimes for model evaluation. The ID regime spans from t_{start} to t_{end} as specified in Table 3, representing the temporal range used for model training and in-distribution testing. The ID-Ext regime extends from t_{start} to $t_{\text{ood.end}}$, combining both the ID temporal range and an extended temporal domain for OOD evaluation. Specifically, for the extended portion beyond t_{end} , we use the state value at t_{end} as the initial condition and continue the integration up to $t_{\text{ood.end}}$, creating a continuous trajectory that tests the model’s extrapolation capability over longer time horizons. All experiments use noise-free data ($\sigma = 0$) with 1000 uniformly sampled time points for each regime. Initial conditions for each system are predetermined and specified in the benchmark configuration rather than randomly sampled, ensuring reproducibility and consistency across all experimental runs.

A.4. State trajectory visualization

We visualize the state trajectories for the benchmark ODE problems generated using the procedure described in the previous subsection. As shown in Figure 9, we present the temporal evolution of state variables for all eight benchmark problems (ID 1-8), where each subfigure displays both the ID training regime (solid lines) and the ID-Ext evaluation regime (extended temporal domain). The visualization demonstrates the diversity of dynamical behaviors in our benchmark suite, ranging from simple polynomial dynamics in the SIR model to complex nonlinear oscillations in systems like CDIMA and the Glider systems.

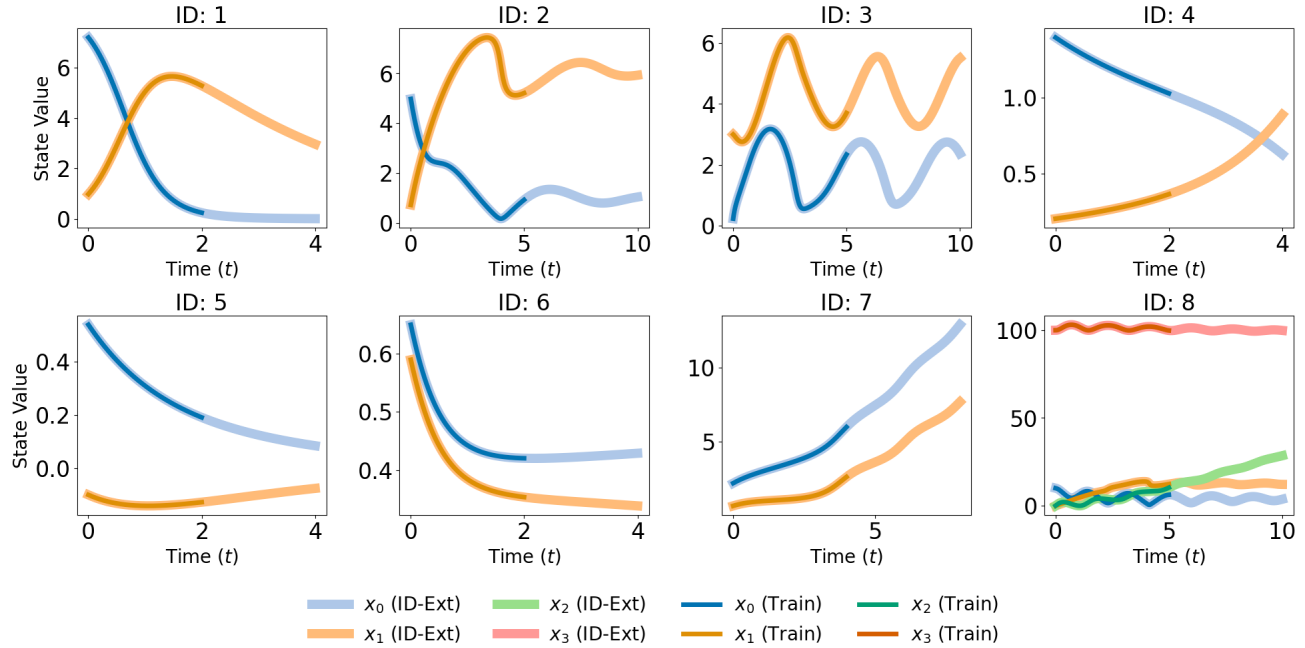


Figure 9. State trajectories for benchmark ODEs in ID and ID-Ext regimes. The figures display: ID 1: SIR infection model; ID 2: Glider (dimensionless); ID 3: Reduced model for chlorine dioxide-iodine-malonic acid reaction; ID 4: Isothermal autocatalytic reaction model; ID 5: Interacting bar magnets; ID 6: Binocular rivalry model; ID 7: Oscillator death model; ID 8: Glider (physical units).

B. Implementation details

B.1. DoLQ hyperparameters

The specific hyperparameters used for the DoLQ framework in our experiments are detailed in Table 5. These settings were chosen to balance exploration capability with computational efficiency.

Table 5. Hyperparameters for DoLQ. Values are derived from the experimental configuration specific to the results reported.

Category	Parameter	Value	Description
LLM Configuration	Sampler Model	gemini-2.5-flash-lite	Model for generating candidate terms
	Scientist Model	gemini-2.5-flash-lite	Model for reasoning and evaluation
	Sampler Temperature	0.9	High temperature for diverse generation
	Scientist Temperature	0.6	Lower temperature for stable reasoning
	Max Tokens (Sampler)	3000	Constraint on generation length
	Max Tokens (Scientist)	3000	Constraint on evaluation length
Search & Optimization	Evolution Iterations	100	Total number of search cycles
	Max Parameters	8	Maximum trainable coefficients per term
	Differential evolution tolerance	10^{-5}	Convergence threshold for differential evolution
	BFGS Tolerance	10^{-9}	Convergence threshold for BFGS
System Settings	Differential evolution strategy	best1bin	Standard differential evolution strategy
	Differential evolution population size	20	Number of candidates in the differential evolution pool
	Forget Probability	0.01	Chance to revive a banned term
	Timeout	240s	Maximum execution time per iteration

In all experiments, the Sampler Agent produces 3 candidate hypotheses per iteration, and each equation in a candidate is constrained to contain at most 10 terms. We do not impose a separate sparsity regularizer; instead, sparsity is controlled operationally through this term cap together with the Scientist Agent’s keep/hold/remove feedback and the subsequent ablation-based filtering. Outputs exceeding the term cap are discarded before optimization. If candidate evaluation produces overflow, underflow, division-by-zero, NaN, or ∞ , we assign the corresponding objective value to ∞ and discard that candidate from further consideration in the current iteration.

B.2. Experimental settings for baseline models

To ensure a fair comparison with our proposed framework, DoLQ, we incorporated the system description $\mathcal{T}$ (providing domain-specific context and physical principles) into the prompts for each baseline method (LASR, LLM-SR, EDL, ICSR), adapted to their respective prompt structures. All baseline methods utilize Gemini 2.5 Flash Lite (Team et al., 2023) as the underlying LLM. For experimental parameters not explicitly mentioned in the following tables, we followed the default settings of each respective baseline method. Specifically for ICSR, the binary and unary operators listed in Table 9 were explicitly provided in the prompt to define the permissible function space.

Table 6. Experimental settings for LASR.

Setting	2D	4D
Description	Included	Included
Iterations	31	31
Populations	6	6
Variable Names	x_0, x_1	x_0, x_1, x_2, x_3
Binary Operators	+, −, *, /, ^	
Unary Operators	cos, sin, log, exp, tan, abs, sinh, cosh, tanh, sign, sqrt	
Use LLM	True	
Use Concepts/Evolution	True	
LLM Probabilities	0.02 (Crossover, Mutate, Randomize)	

Table 7. Experimental settings for LLM-SR.

Setting	2D	4D
Description	Included	Included
Max Samples (Iterations)	100	100
Number of Islands	3	3
Reset Period	483	304
Cluster Sampling Temp. Period	302	190

Table 8. Experimental settings for EDL.

Setting	2D	4D
Description	Included	Included
Max Epoch (Iterations)	100	100
Operands	x_0, x_1	x_0, x_1, x_2, x_3
Binary Operators	+, −, *, /, ^	
Unary Operators	cos, sin, log, exp, tan, abs, sinh, cosh, tanh, sign, sqrt	
Metric	R^2	
Number of Samples (M)	10	
Optimize Type	evolution_optimize	
Mode	nonlinear	

Table 9. Experimental settings for ICSR.

Setting	2D	4D
Description	Included	Included
Iterations	100	100
Operands	x_0, x_1	x_0, x_1, x_2, x_3
Binary Operators (Prompt)	+, −, *, /, ^	
Unary Operators (Prompt)	cos, sin, log, exp, tan, abs, sinh, cosh, tanh, sign, sqrt	

The hyperparameter configurations for each baseline method are summarized in Tables 6, 7, 8, and 9. Table 6 shows the LASR settings, which employ a hybrid evolutionary approach with LLM-guided mutations at a probability of 0.02, running 31 iterations with 6 populations for both 2D and 4D systems. Table 7 details the LLM-SR configuration, which utilizes an island-based evolutionary strategy with 3 islands and adaptive reset mechanisms (reset period of 483 for 2D and 304 for 4D). Table 8 presents the EDL setup, configured with 100 epochs and using R^2 as the fitness metric with 10 samples per generation. Finally, Table 9 specifies the ICSR parameters, which performs 100 iterations of in-context symbolic regression with the operator set explicitly provided in the prompt.

B.3. Algorithm for term retention strategy

This algorithm details the “simple ablation test” mentioned in Section 3.3.1, classifying terms based on their impact on residual MSE.

Algorithm 1: Term Contribution Analysis via Ablation Test

Input: Equation $f(\mathbf{x}; \theta)$, Parameters $\theta = \{c_1, \dots, c_k\}$, Data $\mathcal{D}$, Threshold $\delta = 0.05$
Output: Classification per term (**Good**/**Bad**/**Neutral**)

```

1: Calculate Baseline Error:
    $MSE_{base} \leftarrow \text{Evaluate}(f(\mathbf{x}; \theta), \mathcal{D})$ 
2: For each term  $i \in \{1, \dots, k\}$ :
    $c_{temp} \leftarrow \theta.\text{copy}()$ 
    $c_{temp}[i] \leftarrow 0$  // Temporarily remove term  $i$ 

    $MSE_{ablated} \leftarrow \text{Evaluate}(f(\mathbf{x}; c_{temp}), \mathcal{D})$ 

    $\Delta \leftarrow (MSE_{ablated} - MSE_{base}) / (MSE_{base} + \epsilon)$  // Calculate Change Rate

   If  $\Delta > \delta$ :
       Classification  $\leftarrow$  Good (Positive Impact)
   Else If  $\Delta < -\delta$ :
       Classification  $\leftarrow$  Bad (Negative Impact)
   Else:
       Classification  $\leftarrow$  Neutral
3: Return Classifications
    
```

Decision logic for term retention In the ablation test, we do not re-optimize the remaining coefficients after zeroing a term; instead, we reuse the previously optimized coefficient vector and set only the selected coefficient to zero. We adopted this approximation because a pilot comparison with full re-optimization changed only a very small number of keep/remove decisions while substantially increasing runtime. All optimization and ablation steps are performed on the given training trajectory rather than on a separate validation set.

The final decision for each term is determined by combining the qualitative evaluation from the Scientist LLM (Figure 15) and the quantitative evaluation from the Simple Ablation Test (Algorithm B.3). The decision matrix, including the accumulated hold mechanism, is presented in Table 10. Specifically, we enforce a “2-strike” rule where terms maintained in the *Hold* state for two consecutive iterations are definitively removed (**Remove**) if they fail to improve.

Table 10. Term Retention Decision Matrix. It summarizes the final action determination based on the dual evaluation results and the accumulated hold mechanism.

Semantic Quality (Scientist LLM)	Quantitative Impact (Ablation Test)	Previous State (Iteration $t - 1$)	Final Action (Iteration t)
Bad	Any	Any	Remove
Good	Good (Positive)	Any	Keep
Good	Neutral / Bad	None / Keep	Hold (1st)
		Hold (1st strike)	Hold (2nd)
		Hold (2nd strike)	Remove
Neutral	Any	None / Keep	Hold (1st)
		Hold (1st strike)	Hold (2nd)
		Hold (2nd strike)	Remove

B.4. Function construction

As shown in Figure 13, the Sampler agent generates a list of candidate terms. To enable numerical optimization of the coefficients, we transform these symbolic terms into an executable Python function. Figure 10 illustrates this process, where each term is assigned a learnable parameter (e.g., `params[i]`) and combined into a formatted function body.

Function Construction Example

Input (Term List)

```
x0.t= [
"x0",
"np.sin(x1)",
"x0*x0"
]
```

 $\Rightarrow$

Output (Python Code)

```
def x0.t(x0, x1, x2, x3, params):
import numpy as np
return (params[0]*x0) + \
        (params[1]*np.sin(x1)) + \
        (params[2]*(x0*x0)) + \
        params[3]*1
```

Figure 10. Visual representation of the function construction process. A list of symbolic terms (left) is converted into an executable Python function (right) with learnable coefficients (`params`) and a bias term, enabling numerical optimization.

C. Comprehensive experimental results

C.1. Discovered governing equations

In this section, we present the explicit mathematical forms of the governing equations discovered by each methodology for the benchmark ODE systems, detailing the structure and the optimized high-precision coefficients. These results are systematically organized in Table 11, which provides a comprehensive comparison across all eight benchmark problems (ID 1-8 corresponding to the systems specified in Table 3). For each problem, the table lists the discovered equations from DoLQ and all baseline methods (EDL, ICSR, LASR, LLM-SR), showing the complete equation structure with optimized coefficients for every dimension. This detailed presentation facilitates direct structural comparison between the equations recovered by our framework and those from baseline methods, enabling qualitative assessment of which methods successfully capture the ground truth terms versus those that propose spurious or incomplete dynamics.

Table 11. Governing equations discovered by each methodology.

ID	Method	Dim.	DiscoveredEquations
1	DoLQ(ours)	$\dot{x}_0$	$(-0.4000164923132323 * (x0 * x1)) + (1.1797068684934541e - 06 * (-x0 * x1 * x1)) + 0.00014926603696994873 * 1$
		$\dot{x}_1$	$(0.3999858850733014 * (x0 * x1)) + (0.3133017263306753 * (-x1)) + (-0.00010236009691644724 * (x1 * x1)) - 0.000729280144169156 * 1$
	EDL	$\dot{x}_0$	$-0.4 * x0 * x1$
		$\dot{x}_1$	$-0.1128 * 2 * x0^2 + 0.0051 * 2 * x1^3 + 0.0064 * x0 * x1 + 0.4826 * x1^2 - 0.2024 * x0 * \sqrt{x1}$
	ICSR	$\dot{x}_0$	$-0.4 * x0 * x1 + 0.0001$
		$\dot{x}_1$	$0.4 * x0 * x1 - 0.314 * x1 - 0.0002$
	LASR	$\dot{x}_0$	$x1 * (x0 * -0.4000072832475115)$
		$\dot{x}_1$	$x1 * ((x0 * 0.40001272551202366) + -0.314017558209762)$
	LLM-SR	$\dot{x}_0$	$(-0.5113286722182488 * (x0 * x1) / 1.2782811394075673) - (-1.6210813251309898e - 05 * x1)$
		$\dot{x}_1$	$(0.7940229825361336 / (1 + -0.0007829235554072554 * (x0 + x1))) * (1 / (1 + \exp(-0.0023009370812258584 * ((x1 / (x0 + x1)) - -0.7319833222420157)))) * x0 * x1 - 0.31384689979086494 * x1$
	DoLQ(ours)	$\dot{x}_0$	$(-0.9999342734666715 * (\sin(x1))) + (-0.19999685243662685 * (x0^2)) + (-3.159527567024422e - 05 * (x0 * \sin(x1))) + 8.291723665240451e - 06 * 1$

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Table 11 – Continued from previous page

ID	Method	Dim.	<i>DiscoveredEquations</i>
3	EDL	$\dot{x}_1$	$(-0.9996732126004018 * (\cos(x1)/x0)) + (-6.162357405925185e - 05 * (x0 * \sin(x1))) + (1.0002613154943063 * (x0)) - 0.0006655465341315331 * 1$
		$\dot{x}_0$	$-0.2 * \sin(x1) - 1.0 * x0^2$
		$\dot{x}_1$	$1.2906 * x1 * \log(\text{abs}(x0)) + 3.7156 * x1 + 0.4329 * x0^2 * x1 - 1.2452 * x0 + 1.55 * x1 * \cos(x0)$
	ICSR	$\dot{x}_0$	$-0.2 * x0^2 - 1.0 * \sin(1.0 * x1) - 0.0001$
		$\dot{x}_1$	$-0.5662/(x0^{1.894} * x1^{0.5626}) + 0.8685 * x0 - 0.1773 * x1 + 1.0748$
	LASR	$\dot{x}_0$	$(x0 / -1.7671421572704085) - \sin(x1)$
		$\dot{x}_1$	$x0 - (\cos(x1)/x0)$
	LLM-SR	$\dot{x}_0$	$(-1.387691981374681 * 0.9997683841467897 * \sin(x1) + (-0.27884783934870216 * \text{square}(x0) - (-0.00023524831049328578) * \text{power}(x0, 3)) + ((-0.0005339077881395173) * x0 * \cos(x1) + (-0.0009606900297937713) * x0 * \sin(x1)) + ((1.1025607057532369e - 05) * x0 * \text{square}(x1) + (0.0001392070749430701) * \text{square}(x0) * x1 + (1.0871856110837581e - 05) * x0 * \text{square}(\cos(x1)) + (0.0003566962804791643) * \text{square}(x0) * \sin(x1)))/1.387691981374681$
		$\dot{x}_1$	$((1.00016068611199 * x0^2) + (-0.0003203621386719613 * x0) + (-0.00026883955217234126) - 0.9994481859297301 * \cos(x1))/\text{maximum}(x0, 1e - 6)$
	DoLQ(ours)	$\dot{x}_0$	$(-0.999924337295147 * (x0)) + (0.0013713847096808362 * (x0/(1.0 + x0^2))) + (4.0000051777750265 * (-x1 * x0/(1.0 + x0^2))) + 8.899331099033029 * 1$
		$\dot{x}_1$	$(1.3999732914208531 * (-x0 * x1/(1.0 + x0^2))) + (-4.4947665548779136e - 05 * (-x0^2/(1.0 + x0^2))) + (-1.3999589124979535 * (-x0)) + 4.897208257542713e - 07 * 1$
3	EDL	$\dot{x}_0$	$-7.159 * x1^3 + 10.3164 * x0 - 4.0771 * \cos(x1)^2 - 0.3057 * \text{sqrt}(x0) + 0.0662 * x1$
		$\dot{x}_1$	$-0.7501 * \text{sqrt}(x0) + 0.2166 * x1^2 + 10.8516 * x0^2 - 9.7161 * x1 - 0.3873 * \cos(x0)^3$
	ICSR	$\dot{x}_0$	$-4.0 * x0 * x1/(x0^2 + 0.9999) - 1.0002 * x0 * \tanh(2.0678 * x1) + 8.9004$
		$\dot{x}_1$	$-17.0218 * x0 * \tanh(2.5653 * x0) + 18.7118 * x0 - 3.2262 * x1/(x0 + 3.8981) - 0.6359$
	LASR	$\dot{x}_0$	$((\sin(x0) + -0.7225424108770416) * -1.3716248441728514) + \tan(\sinh(\sin(x1 + -1.271652676453637)))$
		$\dot{x}_1$	$\log(x0)$
	LLM-SR	$\dot{x}_0$	$(7.2413552436477575 * (x0/(-0.09692412820488534 + x0))) * (1 - (x1^{1.6726750850832244})/(4.267112440332994^{1.6726750850832244} + x1^{1.6726750850832244})) + (-1.6934497454181765 * x0 * x1 * ((x0^{-2.2251470718338093})/(1.2897199381103897^{-2.2251470718338093} + x0^{-2.2251470718338093})))$
		$\dot{x}_1$	$(0.15305571645674615 * (x0/(1.4598732465999258 + x0))) * (1 - 17.709388951838832 * x1) - (-2.096703476296624 * ((x0^{1.9666706388834827})/(2.087291118909436^{1.9666706388834827} + x0^{1.9666706388834827})) * x1) - (-1.3593840857406783 * x0)$
	DoLQ(ours)	$\dot{x}_0$	$(0.4973058292805835 * (-x0)) + (0.21746562709380776 * (-x0 * x1)) + (0.6344054886410723 * (-x1 * x1)) + 0.5264411613474004 * 1$

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Table 11 – Continued from previous page

ID	Method	Dim.	<i>DiscoveredEquations</i>
5	EDL	$\dot{x}_1$	$(0.41858642320557227 * (x_0 * x_1^3)) + (-1.6764582281044875 * (x_1/(1 + x_1))) + (1.5117151384013812 * (\tanh(x_1))) + 0.028354488818915925 * 1$
		$\dot{x}_0$	$-0.1566 * \sin(x_1)^2 - 1.404 * x_0^2 + 0.5305 * x_1$
		$\dot{x}_1$	$0.8039 * x_0 * x_1 + 0.0724 * \sin(x_1)^2$
	ICSR	$\dot{x}_0$	$2.4649 * x_0 * \cos(0.9101 * x_1) - 2.9665 * x_0 + 0.5033$
		$\dot{x}_1$	$0.0692 * x_0 + 0.6177 * x_1 - 0.1679$
	LASR	$\dot{x}_0$	$\text{abs}(\sinh(\tanh(\sin(\tan(x_1 + ((x_0^{0.22334073977903218}) + (x_0/x_0))) + x_0))) + (x_0 * 0.3514105571067361)) * -0.35812758021702057$
		$\dot{x}_1$	$((x_0 * x_1) - 0.019991627704281027) * x_1$
	LLM-SR	$\dot{x}_0$	$(0.7575404568207408 * (x_0^{1.0100603887529287}) * (x_1^{0.6955603195414775})) + (-0.6206768205830506 * x_0) + 0.39548215725015795 + (-1.6012562268622113 * x_1) + (0.5863974471340245 * (x_1^{0.698948044530037}))$
		$\dot{x}_1$	$(0.007339212327735855 * x_0) + (0.9624797409788882 * x_1^2) + (0.7873542337819555 * x_1^3) - (-0.06037485229609352 * x_1) - (-0.06037485265191514 * x_1) - (0.09930729898681726 * x_0 * x_1) - (1.2126452269944556 * x_1^3) + (0.0073391823827276045 * x_0)$
	DoLQ(ours)	$\dot{x}_0$	$(-1.0015488038824854 * (\sin(x_0))) + (0.33102674079192346 * (\sin(x_0 - x_1))) + (-0.0018115741826902861 * (\cos(x_0))) + 0.0017490917848575025 * 1$
		$\dot{x}_1$	$(-0.9981457993630254 * (\sin(x_1))) + (0.3300067495528232 * (\sin(x_1 - x_0))) + (-0.012578074617818194 * (\cos(x_1))) + 0.012716069769122665 * 1$
	EDL	$\dot{x}_0$	$-0.3375 * \sin(x_0) - 0.684 * x_1$
		$\dot{x}_1$	$-0.2973 * x_1 - 0.6309 * x_0$
	ICSR	$\dot{x}_0$	$5.0117 * x_0 * \sin(x_1) - 2.2897 * x_1 - 0.2639$
		$\dot{x}_1$	$-0.2981 * x_0 - 0.6191 * x_1 + 0.0019$
	LASR	$\dot{x}_0$	$(x_0 * -0.66306757779156) + 0.04301494511373006$
		$\dot{x}_1$	$x_0 * ((x_0 / -1.192434344729412) + 0.25749851031674253)$
	LLM-SR	$\dot{x}_0$	$-1.000043264361751 * \sin(x_0) + -0.3300718799861972 * \sin(x_1 - x_0) + -1.568587192043395e - 05 * \cos(x_1 + x_0)$
		$\dot{x}_1$	$-0.9938291044117465 * x_1 + -0.3292639827583344 * \sin(x_0 - x_1) - (-2.656564954764913e - 06 / (x_0^2 + x_1^2 + 1e - 9)^{1.5}) + -0.0003613964745632571 * x_0 * \sin(x_1)$
6	DoLQ(ours)	$\dot{x}_0$	$(1.3286098112316618 * (-x_0)) + (1.178235675777727 * (-x_1)) + (-0.6010789351853646 * (-\text{power}(x_0, 3))) + 0.9285456156774677 * 1$
		$\dot{x}_1$	$(0.23753405260404614 * (-x_1)) + (-0.7898637222390873 * (-x_0)) + (-0.8360904268130745 * (\tanh(x_1))) + (2.290634305077196 * (-\tanh(x_0))) + 0.9338877404981648 * 1$
	EDL	$\dot{x}_0$	$-1.1878 * x_1 - 0.467 * \log(\sin(x_0))$
		$\dot{x}_1$	$-1.1299 * \log(\sin(x_0)) - 0.4331 * x_1$
	ICSR	$\dot{x}_0$	$-1.8485 * x_1 - 0.1948 * \sin(x_0) + 0.7327$

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Table 11 – Continued from previous page

ID	Method	Dim.	<i>DiscoveredEquations</i>
7	LASR	$\dot{x}_1$	$0.2393 * x_0 - 2.1268 * x_1 + 0.6431$
		$\dot{x}_0$	$\cos(\exp(\tan(x_0)))$
		$\dot{x}_1$	$\tanh(0.14631324378767832/x_1) - x_0$
	LLM-SR	$\dot{x}_0$	$((-2.249593853293885 * (x_0 / (1.1957556897109163 + x_0)) + 1.0667454050960814) - (3.6741828960616694 * (x_1 / (2.3342324749372967 + x_1))))$
		$\dot{x}_1$	$x_1 * (-2.9587039457791064 * (1 - (x_1 / 0.5623115553288086)) - (1.966298296973735 * x_0 * 1.966298237331697)) + 0.9508865699097846$
	DoLQ(ours)	$\dot{x}_0$	$(-1.1655257292555141 * (\sin(x_1 - x_0))) + (1.22475958770386 * (\cos(x_0))) + (-0.1090013166633977 * (x_0 * \sin(x_0))) + 0.817067905629052 * 1$
		$\dot{x}_1$	$(-2.2117156170501504 * (x_1)) + (-1.291170751585586 * (x_0)) + (-4.676614135451172 * (\cos(x_1))) + (1.8750516943971047 * (\sin(x_1 - x_0))) + 10.469917369126234 * 1$
	EDL	$\dot{x}_0$	$-0.817 * x_1^2 - 4.3577 * x_0^3 * \sin(x_0) + 2.7509 * x_1 + 1.5327 * x_0 * \sqrt{x_1} + 0.0552 * x_0 * \cos(x_0)$
		$\dot{x}_1$	$-1.7872 * x_0 + 0.3915 * \sqrt{x_1} - 0.7313 * \cos(x_0) + 0.1632 * x_0^2 * x_1 + 0.1673 * x_1^3$
	ICSR	$\dot{x}_0$	$-0.2806 * x_0 + 0.6592 * x_1 * \cos(1.1398 * x_0) + 2.0662$
		$\dot{x}_1$	$-0.2806 * x_0 + 0.6592 * x_1 * \cos(1.1398 * x_0) + 1.6062$
	LASR	$\dot{x}_0$	$(\cos(x_0) * \sin(x_1)) + 1.4320121866725906$
		$\dot{x}_1$	$(\cos(x_0) * \sin(x_1)) + 0.9720121867249$
8	LLM-SR	$\dot{x}_0$	$(-(-1.231629475981958) * x_0 - 2.158066362365913 * x_0^2 - (-0.3037486836573964) * x_0^3) + (1.3169135770056555 * x_1 + 2.287561908522607 * x_1^2 - 1.2108364457957292 * x_1^3) + (-0.8950111650180933 * (x_1 - x_0) + 1.609022512581006 * (x_1 - x_0)^2 + 0.3125475185298518 * (x_1 - x_0)^3)$
		$\dot{x}_1$	$(0.14605034843290507 * x_1) + (0.3954492324348049 * x_0) + (1.249712176091898 * (x_0 - x_1)) + (-0.2869847813397451 * x_1^3) + (0.03189090914482745 * x_0^3) + (-0.08031202424973191 * x_1^2) + (-0.5242674943108351 * x_0^2) + (0.017802263147943736 * x_0^4) + (-0.15007455215067977 * x_1^4) + (-0.09580817521497335 * (x_0 - x_1)^3)$
	DoLQ(ours)	$\dot{x}_0$	$(-9.809127338404517 * (\sin(x_1))) + (-9.411027230248375e - 05 * (\cos(x_1))) + (-0.030620460686854464 * (x_0 * x_0)) - 4.9118855130869645e - 05 * 1$
		$\dot{x}_1$	$(0.9990536032782628 * (-9.81/x_0 * \cos(x_1))) + (0.6130165352306062 * (x_0)) + (-0.0032944410806144153 * (\cos(x_1))) - 0.0023916253783730504 * 1$
		$\dot{x}_2$	$(1.0000423141055985 * (x_0 * \cos(x_1))) + (-1.5764831178241946e - 05 * (x_0 * x_0 * \cos(x_1))) + 0.00014446800549605261 * 1$
		$\dot{x}_3$	$(1.0000402047694594 * (x_0 * \sin(x_1))) + (-2.003183912960774e - 05 * (x_0 * x_0 * \sin(x_1))) + (-6.303122694043895e - 06 * (9.81 * \cos(x_1))) - 4.353532868248261e - 05 * 1$
	EDL	$\dot{x}_0$	$-0.0306 * \sin(x_1) - 9.8091 * x_0^2$
		$\dot{x}_1$	$-0.9504 * x_0^3 - 0.2996 * x_1^2 - 0.0006 * x_3 + 1.3296 * \sqrt{\text{abs}(x_1)} - 175.3583 * x_0/2$
		$\dot{x}_2$	$0.9999 * x_0 * \cos(x_1)$
		$\dot{x}_3$	$0.9999 * x_0 * \sin(x_1)$

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Table 11 – Continued from previous page

ID	Method	Dim.	DiscoveredEquations
ICSR		$\dot{x}_0$	$-0.2813 * x_0 + 0.0015 * x_1 * x_2 - 0.0968 * x_3 * \sin(1.0007 * x_1) + 0.4387$
		$\dot{x}_1$	$-0.0605 * x_0 - 6.8291 * x_1 * \exp(-2.7884 * x_0) - 0.0042 * x_2 * x_3 + 5.306$
		$\dot{x}_2$	$0.9998 * x_0 * \cos(x_1) + 0.0217$
		$\dot{x}_3$	$0.9999 * x_0 * \sin(x_1) + 0.0002$
LASR		$\dot{x}_0$	$\sin(x_1) * -9.813480254368727$
		$\dot{x}_1$	$((\log(x_0) - ((x_2 - 5.963031610926088) * 0.24925925160994267)) * 8.24239810331568)/x_0$
		$\dot{x}_2$	$\cos(x_1) * x_0$
		$\dot{x}_3$	$\sin(x_1) * x_0$
LLM-SR		$\dot{x}_0$	$(-9.807899351331447 * \sin(x_1)) + -((0.4145037567137417 + 1.5751662511703768 * (-0.0001796563617646803 + -8.696424882717339e - 05 * x_1)^2) * 0.07394440250282637 * x_0^2) + ((-0.0001796563617646803 + -8.696424882717339e - 05 * x_1) * 0.07394440250282637 * x_0^2 * \sin(x_1)) + (-0.00026688377456063225 * x_0) + (3.1382118431267775e - 05 * x_2) + (-7.49115792450938e - 06 * x_3) + (0.00024089007421140555 * x_0 * \cos(x_1) * \sin(x_1))$
		$\dot{x}_1$	$((0.4312345753897334 * x_0) + (-7.190875450891451 * x_0^2) + (115.22337211688654 * \cos(x_1))) / (-11.719230243958577 * x_0 + -0.023905720212081497 + 1e - 9) + (0.0004499464651382155 * x_2) + (0.0002873146658744879 * x_3) + (3.203979516768474e - 05 * x_0 * x_1)$
		$\dot{x}_2$	$(-(-8.313688998942643e - 05) * \sin(x_1)) + (0.00012926385963329574 * -0.0002110727122192323 * x_3 * x_0^2 * \sin(x_1)) + (-0.00012926385963329574 * 0.05 * x_0^2 * \cos(x_1)) + (-0.00012926385963329574 * 1.5157577205870516 * (-0.0002110727122192323 * x_3)^2 * x_0^2 * \cos(x_1)) + (-1.0 * \exp(-(x_3 - 1.0)/1.0)^2) * x_0^2 * \cos(x_1)) + (-0.9999727429727798 * (x_0 - 4.820794712947023e - 05) * \cos(x_1)) + (1.7829345690512744e - 08 * x_3 * x_1 * x_0)$
		$\dot{x}_3$	$(0.00013369546648802163 * x_0 * \cos(x_1)) + (-6.586738720708613e - 06 * x_0^2 * \sin(x_1)) + (-4.30440238524879e - 06 * x_0^3 * \cos(x_1)) + (-2.158359661264577e - 05 * x_2 * x_0 * \cos(x_1)) + (0.00011240049424943671 * x_0) + (-0.0001459955779485213 * x_0 * \sin(x_1)^2) + (1.2061925694746197e - 05 * x_0^2 * \cos(x_1)) + (-3.152838841683395e - 05 * x_0 * \sin(x_1) * \cos(x_1)) + 2.7038601280307466e - 05 + (0.9999723594741682 * x_0 * \sin(x_1))$

C.2. Quantitative results

To quantitatively assess the fidelity of the discovered models, we employ the Normalized Mean Squared Error (NMSE) metric, evaluating the deviation between the ground truth dynamics and the identified governing equations for each state variable dimension. Table 12 presents these NMSE values across all benchmark problems (ID 1-8) for both DoLQ and baseline methods. The table is structured to show four evaluation metrics for each dimension: ID NMSE (Residual) and ID NMSE (Integral) measure performance on the training domain, while ID-Ext NMSE (Residual) and ID-Ext NMSE (Integral) assess generalization to extended temporal regimes. Residual NMSE evaluates point-wise derivative matching, whereas Integral NMSE measures trajectory prediction accuracy through numerical integration from initial conditions. This comprehensive quantitative comparison enables systematic analysis of both fitting quality and extrapolation capability across different problem complexities and functional forms.

Let $\hat{x}_{j,i}$ denote the derivative predicted by the discovered equation at time index i , and let $\tilde{x}_{j,i}$ denote the trajectory obtained by numerically integrating the discovered ODE from the initial condition. The two scale-normalized metrics are defined as follows:

$$\text{Residual NMSE}_j = \frac{\sum_i (\dot{x}_{j,i} - \hat{x}_{j,i})^2}{\sum_i \dot{x}_{j,i}^2 + \epsilon}$$

$$\text{Integral NMSE}_j = \frac{\sum_i (x_{j,i} - \tilde{x}_{j,i})^2}{\sum_i x_{j,i}^2 + \epsilon}$$

Residual MSE in Eq. (2) is used inside the search loop for parameter optimization and term ablation, whereas residual and integral NMSE are used only for final scale-normalized evaluation.

Table 12. NMSE comparison for each dimension.

ID	Method	Dim	ID NMSE (Residual)	ID NMSE (Integral)	ID-Ext NMSE (Residual)	ID-Ext NMSE (Integral)
			Value	Value	Value	Value
1	DoLQ(ours)	x_0	2.13×10^{-8}	3.45×10^{-9}	1.04×10^{-8}	2.36×10^{-9}
		x_1	1.28×10^{-8}	4.52×10^{-9}	2.07×10^{-8}	5.03×10^{-8}
	EDL	x_0	2.39×10^{-8}	—	9.16×10^{-9}	—
		x_1	3.71×10^1	—	2.80×10^1	—
	ICSR	x_0	2.92×10^{-8}	1.89×10^{-9}	1.22×10^{-8}	1.32×10^{-9}
		x_1	2.69×10^{-8}	7.36×10^{-9}	1.62×10^{-8}	2.24×10^{-8}
	LASR	x_0	2.22×10^{-8}	5.62×10^{-9}	8.56×10^{-9}	3.42×10^{-9}
		x_1	1.34×10^{-8}	8.77×10^{-9}	6.64×10^{-9}	1.38×10^{-8}
	LLM-SR	x_0	2.12×10^{-8}	3.70×10^{-9}	8.59×10^{-9}	2.21×10^{-9}
		x_1	1.19×10^{-8}	4.58×10^{-9}	2.37×10^{-8}	5.36×10^{-8}
	DoLQ(ours)	x_0	4.93×10^{-9}	1.42×10^{-8}	5.83×10^{-9}	2.19×10^{-8}
		x_1	3.51×10^{-8}	4.69×10^{-8}	3.64×10^{-8}	5.06×10^{-8}
2	EDL	x_0	1.43×10^1	1.02×10^0	1.19×10^1	1.43×10^0
		x_1	1.64×10^2	—	3.19×10^2	—
	ICSR	x_0	1.21×10^{-8}	0.000110	1.51×10^{-8}	0.0618
		x_1	0.005294	0.000763	0.0772	0.0222
	LASR	x_0	0.1127	0.0653	0.1478	0.0721
		x_1	5.61×10^{-8}	0.1116	5.33×10^{-8}	0.1431
	LLM-SR	x_0	1.41×10^0	9.32×10^0	1.15×10^0	1.80×10^1
		x_1	1.19×10^0	8.20×10^0	1.12×10^0	1.59×10^1
	DoLQ(ours)	x_0	4.22×10^{-8}	3.79×10^{-8}	3.65×10^{-8}	1.16×10^{-7}
		x_1	3.52×10^{-9}	1.58×10^{-8}	1.06×10^{-8}	1.14×10^{-7}
	EDL	x_0	2.14×10^5	—	2.65×10^5	—
		x_1	4.31×10^2	—	4.15×10^2	—
3	ICSR	x_0	5.11×10^{-8}	4.03×10^{-5}	5.11×10^{-8}	0.000210
		x_1	0.001031	8.09×10^{-5}	0.000756	0.000303
	LASR	x_0	0.0688	1.66×10^1	0.0758	3.08×10^1
		x_1	0.5129	6.14×10^0	0.5047	6.62×10^1
	LLM-SR	x_0	0.005255	0.004934	0.006184	0.007153
		x_1	8.93×10^{-5}	0.005235	9.19×10^{-5}	0.0116
	DoLQ(ours)	x_0	1.92×10^{-6}	3.27×10^{-9}	0.2675	0.000975
		x_1	9.95×10^{-8}	1.71×10^{-9}	0.001386	5.48×10^{-5}
	EDL	x_0	3.05×10^3	2.94×10^1	1.03×10^3	5.83×10^0
		x_1	7.11×10^1	9.47×10^0	1.92×10^0	2.35×10^0
	ICSR	x_0	7.54×10^{-6}	1.35×10^{-6}	0.0120	0.001100
		x_1	0.000154	1.88×10^{-5}	0.0267	0.007869
4	LASR	x_0	0.000159	6.88×10^{-7}	1.05×10^0	0.007504
		x_1	1.19×10^{-7}	2.29×10^{-8}	6.71×10^{-8}	0.000871

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Table 12 – Continued from previous page

ID	Method	Dim	ID NMSE (Residual)	ID NMSE (Integral)	ID-Ext NMSE (Residual)	ID-Ext NMSE (Integral)
			Value	Value	Value	Value
5	LLM-SR	x_0	6.08×10^{-6}	2.45×10^{-8}	0.5790	0.004555
		x_1	7.52×10^{-7}	2.13×10^{-8}	0.007890	0.000784
	DoLQ(ours)	x_0	2.43×10^{-7}	5.61×10^{-10}	1.27×10^{-7}	2.87×10^{-9}
		x_1	4.85×10^{-7}	6.86×10^{-8}	4.56×10^{-7}	1.97×10^{-7}
	EDL	x_0	5.86×10^0	7.16×10^0	2.68×10^0	2.61×10^1
		x_1	2.08×10^1	6.26×10^2	1.56×10^1	8.86×10^2
	ICSR	x_0	0.002610	4.30×10^{-5}	0.2313	0.0435
		x_1	2.58×10^{-6}	1.89×10^{-5}	5.54×10^{-5}	0.0310
	LASR	x_0	0.000346	1.33×10^{-5}	0.0117	0.001680
		x_1	0.009038	0.0114	0.0339	0.0892
	LLM-SR	x_0	2.48×10^{-7}	6.19×10^{-10}	1.17×10^{-7}	4.77×10^{-8}
		x_1	4.91×10^{-7}	6.40×10^{-8}	0.000219	0.000103
6	DoLQ(ours)	x_0	1.10×10^{-6}	3.28×10^{-8}	2.33×10^{-6}	7.31×10^{-8}
		x_1	1.07×10^{-6}	3.71×10^{-8}	7.50×10^{-6}	4.55×10^{-6}
	EDL	x_0	1.11×10^{-5}	—	1.19×10^{-5}	—
		x_1	5.01×10^1	—	7.15×10^1	—
	ICSR	x_0	0.000298	1.76×10^{-5}	0.004846	7.17×10^{-5}
		x_1	0.000426	3.68×10^{-5}	0.0170	0.006141
	LASR	x_0	0.0366	0.0161	0.0281	0.0158
		x_1	0.0185	0.0180	0.0262	0.0620
	LLM-SR	x_0	7.03×10^{-6}	6.29×10^{-7}	5.25×10^{-6}	7.12×10^{-5}
		x_1	2.47×10^{-6}	4.72×10^{-7}	0.000166	0.000228
	DoLQ(ours)	x_0	0.000467	3.62×10^{-5}	6.53×10^0	0.3966
		x_1	0.000239	0.000125	3.70×10^2	0.5569
7	EDL	x_0	3.30×10^5	—	1.56×10^7	—
		x_1	3.93×10^1	—	2.10×10^4	—
	ICSR	x_0	0.000510	2.55×10^{-5}	1.30×10^1	0.4600
		x_1	0.000510	0.000116	1.30×10^1	1.01×10^0
	LASR	x_0	1.20×10^{-7}	3.14×10^{-8}	4.15×10^{-7}	1.79×10^{-8}
		x_1	1.20×10^{-7}	1.43×10^{-7}	4.15×10^{-7}	3.93×10^{-8}
	LLM-SR	x_0	9.64×10^{-6}	8.79×10^{-8}	1.67×10^3	—
		x_1	9.78×10^{-6}	3.77×10^{-7}	5.84×10^3	—
	DoLQ(ours)	x_0	7.23×10^{-8}	1.10×10^{-6}	6.82×10^{-8}	3.94×10^{-6}
		x_1	4.78×10^{-6}	2.01×10^{-7}	3.20×10^{-6}	3.38×10^{-7}
		x_2	1.23×10^{-8}	3.40×10^{-6}	1.68×10^{-8}	6.47×10^{-6}
		x_3	9.38×10^{-9}	1.43×10^{-6}	3.13×10^{-8}	1.01×10^{-5}
	EDL	x_0	2.69×10^3	—	2.60×10^3	—
		x_1	5.67×10^4	—	5.20×10^4	—
		x_2	1.61×10^{-8}	—	2.91×10^{-8}	—
		x_3	1.08×10^{-8}	—	3.22×10^{-8}	—
8	ICSR	x_0	0.000678	1.01×10^0	0.000945	2.85×10^0
		x_1	0.1014	1.14×10^0	1.02×10^0	2.74×10^0
		x_2	3.72×10^{-5}	0.1605	6.03×10^{-5}	0.4885
		x_3	1.52×10^{-8}	2.32×10^0	3.65×10^{-8}	6.67×10^0

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Table 12 – Continued from previous page

ID	Method	Dim	ID NMSE (Residual)	ID NMSE (Integral)	ID-Ext NMSE (Residual)	ID-Ext NMSE (Integral)
			Value	Value	Value	Value
	LASR	x_0	0.0264	4.24×10^0	0.0255	1.64×10^2
		x_1	0.0735	1.31×10^0	1.95×10^0	3.39×10^0
		x_2	1.84×10^{-8}	1.84×10^0	2.05×10^{-8}	0.8254
		x_3	1.51×10^{-8}	6.70×10^0	3.40×10^{-8}	3.85×10^3
	LLM-SR	x_0	6.89×10^{-8}	7.58×10^{-7}	7.08×10^{-8}	1.01×10^{-5}
		x_1	3.89×10^{-6}	1.92×10^{-7}	3.24×10^{-6}	5.18×10^{-7}
		x_2	1.29×10^{-8}	2.81×10^{-6}	1.71×10^{-8}	2.65×10^{-6}
		x_3	9.91×10^{-9}	7.05×10^{-7}	8.81×10^{-8}	5.76×10^{-6}

C.3. Scores

In this section, we summarize the comprehensive evaluation scores for each benchmark problem. Figure 4 and Table 13 present a binary success/failure assessment across all eight problems (P1-P8, corresponding to benchmark IDs 1-8 from Table 3) for each method. The evaluation is based on two distinct criteria: the NMSE test row indicates whether the normalized mean squared error satisfies the designated convergence threshold (integral NMSE $< 10^{-3}$ across all dimensions), and the Term test row verifies whether the discovered equation accurately recovers the correct symbolic terms of the ground truth dynamics after excluding terms with negligible impact. A blue checkmark (✓) denotes success for that problem, while an empty cell indicates failure. The rightmost Total column sums the number of successful problems for each method and evaluation criterion, providing an aggregate view of discovery performance across the benchmark suite.

As prior ODE-discovery baselines do not report structural recovery under a unified false-positive/false-negative (FP/FN) protocol, precision/recall metrics are omitted from the main comparison. Instead, the Term test evaluates whether the recovered equation matches the ground-truth term set after excluding negligible terms, which serves as a strict structure-recovery criterion on benchmarks with known governing equations.

Table 13. Detailed scores for each problem by evaluation type and method.

EVALUATION	METHOD	P1	P2	P3	P4	P5	P6	P7	P8	TOTAL
NMSE TEST	DoLQ	✓	✓	✓	✓	✓	✓		✓	7
NMSE TEST	EDL									0
NMSE TEST	ICSR	✓		✓						2
NMSE TEST	LASR	✓						✓		2
NMSE TEST	LLM-SR	✓				✓	✓		✓	4
TERM TEST	DoLQ	✓	✓	✓		✓			✓	5
TERM TEST	EDL									0
TERM TEST	ICSR									0
TERM TEST	LASR	✓						✓		2
TERM TEST	LLM-SR		✓			✓				2

C.4. Token usage details

This section presents a detailed comparison of token usage between the proposed framework and baseline models. The token consumption data in the following table is reported in thousands (K) of tokens, distinguishing between input tokens (In) and output tokens (Out). For each benchmark problem (ID 1-8), we track two metrics: “Found” columns show the cumulative tokens consumed up to the point when the method successfully discovered a reasonable equation (if applicable), while “Total” columns report the tokens used throughout the entire experimental run until completion or timeout. This distinction enables analysis of both the efficiency of successful discovery and the overall computational cost. All values are reported

directly from the experiment logs collected during our experiments. Note that for DoLQ, the reported token counts represent the combined usage of both the Sampler Agent and the Scientist Agent, and no reasoning tokens were generated during our experiments.

Table 14. Token usage until discovery.

ID	Method	Found (K)		Total (K)	
		In	Out	In	Out
1	DoLQ(ours)	20.2	10.1	226	96.8
	EDL	4.99	6.96	551	439
	ICSR	243	24.8	243	24.8
	LASR	357	387	357	387
	LLM-SR	338	443	338	443
2	DoLQ(ours)	69.6	40.0	289	156
	EDL	149	42.5	465	118
	ICSR	262	39.0	262	39.0
	LASR	426	426	426	426
	LLM-SR	533	602	703	816
3	DoLQ(ours)	188	110	282	166
	EDL	129	302	296	296
	ICSR	309	44.1	309	44.1
	LASR	483	447	483	447
	LLM-SR	652	692	813	813
4	DoLQ(ours)	44.6	26.4	201	109
	EDL	1.49	2.97	193	239
	ICSR	260	35.1	260	35.1
	LASR	401	404	401	404
	LLM-SR	781	775	781	775
5	DoLQ(ours)	20.4	12.4	223	106
	EDL	0.61	0.82	186	222
	ICSR	256	31.1	256	31.1
	LASR	368	420	368	420
	LLM-SR	624	784	624	784
6	DoLQ(ours)	10.3	8.08	239	136
	EDL	0.41	0.53	190	233
	ICSR	255	29.4	255	29.4
	LASR	345	394	345	394
	LLM-SR	730	701	730	701
7	DoLQ(ours)	183	89.4	233	114
	EDL	26.9	29.8	279	287
	ICSR	256	33.9	256	33.9
	LASR	383	407	383	407
	LLM-SR	758	854	758	854
8	DoLQ(ours)	173	68.7	784	264
	EDL	73.6	22.9	989	265
	ICSR	772	72.9	772	72.9
	LASR	809	930	809	930
	LLM-SR	1238	1379	1852	1997

D. Analysis of optimizer adoption

To further analyze the behavior of the hybrid optimizer, we investigate the adoption frequency of BFGS and differential evolution across different problem types. As shown in Figure 11, the selection rate of these methods varies significantly depending on the mathematical complexity of the system. For systems with simpler functional forms, BFGS is frequently sufficient to find the optimal parameters due to the relatively smooth loss landscapes. Conversely, for systems with complex functional forms (e.g., Glider), differential evolution is adopted more frequently. This demonstrates that the global search capability of differential evolution is critical for escaping local minima and accurately identifying parameters in rugged loss landscapes where local gradient-based methods often fail.

Moreover, the left panel of Figure 11 reveals a critical advantage of the hybrid optimizer. When examining $\dot{x}_0$ with the ground truth skeleton structure, we observe that differential evolution achieves lower MSE compared to using BFGS alone. This is particularly significant because assigning better scores to correct skeleton structures provides a crucial advantage during the symbolic regression search process, enabling more effective exploration and convergence to the true underlying dynamics.

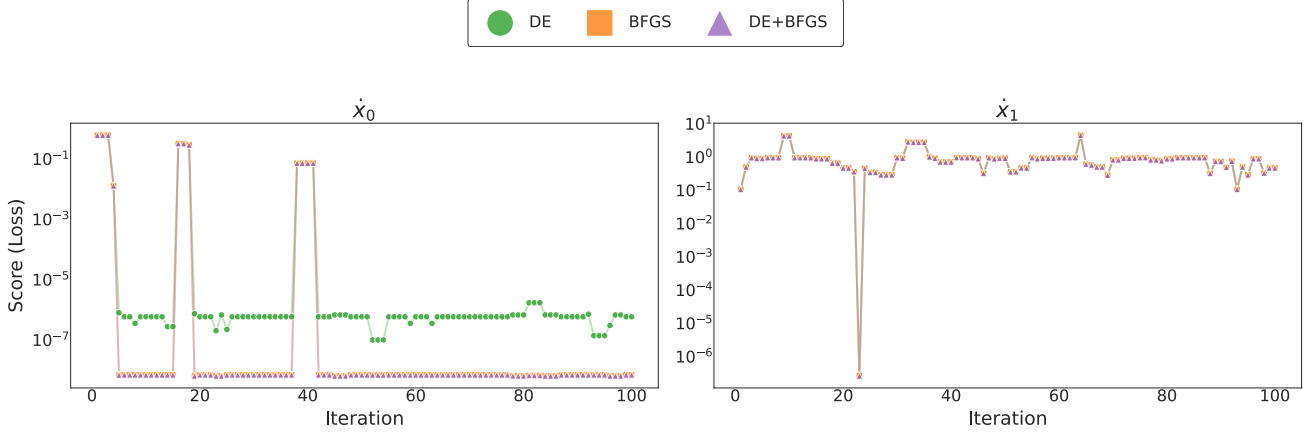


Figure 11. Adoption frequency of BFGS and differential evolution across different ODE systems. Differential evolution is more frequently selected for systems with intricate functional forms where the loss landscape is more rugged, while BFGS is often sufficient for simpler systems.

E. Quantitative results under shifted initial conditions

To evaluate robustness against initial-state variations, the discovered equations are re-assessed using an alternative initial condition (ic_1), maintaining the original ID and ID-Ext temporal regimes and NMSE metrics. The results are summarized in Table 15.

Table 15. Quantitative performance comparison measured by NMSE under shifted initial conditions. We report dimension-averaged results for the same discovered equations, re-evaluated from an alternative initial condition (ic_1) rather than the default initial state, demonstrating robustness and generalization across initial conditions. Bold with underline indicates the best, bold indicates the second best. NaN indicates solver failure or numerical instability.

Benchmark	Metric	Regime	Model				
			ICSR	LASR	LLM-SR	EDL	DoLQ(ours)
SIR(2D)	Residual	ID	<u>1.47e-8</u>	<u>1.41e-8</u>	2.18e-4	1.23e2	1.35e-6
		ID-Ext	<u>1.22e-9</u>	<u>3.30e-9</u>	2.97e-3	NaN	7.15e-5
	Integral	ID	<u>1.44e-8</u>	<u>1.39e-8</u>	2.09e-4	1.29e2	1.37e-6
		ID-Ext	<u>1.13e-9</u>	<u>3.67e-9</u>	8.02e-4	NaN	3.57e-5
CDIMA(2D)	Residual	ID	<u>4.53e-2</u>	4.35e-1	4.80e2	6.10e5	<u>2.32e-8</u>
		ID-Ext	1.12e-1	NaN	<u>6.89e-3</u>	NaN	<u>3.41e-8</u>
	Integral	ID	2.60e-2	4.04e-1	<u>2.42e-2</u>	3.67e5	<u>1.78e-8</u>
		ID-Ext	1.26e-1	NaN	<u>1.59e-2</u>	NaN	<u>9.22e-8</u>
Glider(4D)	Residual	ID	7.12e0	6.17e0	<u>2.44e-4</u>	1.17e5	<u>1.05e-6</u>
		ID-Ext	1.18e3	9.13e2	<u>7.71e-4</u>	NaN	<u>5.79e-6</u>
	Integral	ID	7.43e0	5.90e1	<u>2.76e-4</u>	1.99e5	<u>1.85e-6</u>
		ID-Ext	3.44e3	1.94e3	<u>7.22e-4</u>	NaN	<u>7.49e-6</u>

F. Additional comparison on the 2D dimensionless Glider system (ID 2)

A detailed analysis of the 2D dimensionless Glider system (ID 2) is provided to supplement the main evaluation. Table 16 reports per-dimension NMSE values, Table 17 summarizes token consumption, and Table 18 lists the discovering equations across varying LLM backbones, facilitating a comprehensive comparison of accuracy, efficiency, and symbolic structure.

Table 16. Per-dimension NMSE on the 2D dimensionless Glider system (ID 2) when DoLQ is executed with different LLM models. All settings are identical to Table 5, with only the model changed. The table reports residual and integral NMSE on the ID and ID-Ext evaluation regimes for each state dimension.

Model	Dim	Residual		Integral	
		ID	ID-Ext	ID	ID-Ext
Gemini 2.5 Flash-Lite	x_0	4.93×10^{-9}	5.83×10^{-9}	1.42×10^{-8}	2.19×10^{-8}
	x_1	3.51×10^{-8}	3.64×10^{-8}	4.69×10^{-8}	5.06×10^{-8}
GPT-4o mini	x_0	5.16×10^{-9}	5.07×10^{-8}	1.67×10^{-3}	8.87×10^{-1}
	x_1	4.04×10^{-2}	8.48×10^{-1}	7.75×10^{-3}	3.28×10^{-1}
DeepSeek-V3.2	x_0	4.01×10^{-9}	6.93×10^{-8}	1.33×10^{-8}	6.26×10^{-7}
	x_1	2.85×10^{-8}	5.66×10^{-7}	1.78×10^{-8}	3.22×10^{-7}
Grok 4.1 Fast	x_0	4.04×10^{-9}	6.88×10^{-8}	1.40×10^{-8}	9.11×10^{-7}
	x_1	2.85×10^{-8}	7.77×10^{-7}	1.79×10^{-8}	4.24×10^{-7}

Table 17. Token usage on the 2D dimensionless Glider system (ID 2) when DoLQ is executed with different LLM models. All settings are identical to Table 5, with only the model changed. We report the total input and output tokens consumed during symbolic ODE discovery.

Model	Total input tokens	Total output tokens
Gemini 2.5 Flash-Lite	289K	156K
GPT-4o mini	259K	89K
DeepSeek-V3.2	258K	100K
Grok 4.1 Fast	337K	725K

Table 18. Final equations discovered on the 2D dimensionless Glider system (ID 2) when DoLQ is executed with different LLM models. All settings are identical to Table 5, with only the model changed. For each model, we list the identified governing equations for $\dot{x}_0$ and $\dot{x}_1$.

Model	Dim	Equation
Gemini 2.5 Flash-Lite	$\dot{x}_0$	$-0.999934 \sin(x_1) - 0.199969x_0^2 - 3.160 \times 10^{-5}(x_0 \sin(x_1)) + 8.292 \times 10^{-6}$
	$\dot{x}_1$	$-0.999673 \frac{\cos(x_1)}{x_0} - 6.162 \times 10^{-5}(x_0 \sin(x_1)) + 1.000261x_0 - 6.655 \times 10^{-4}$
GPT-4o mini	$\dot{x}_0$	$-4.597 \times 10^{-5} \cos(x_1) - 0.200003x_0^2 - 0.999975 \sin(x_1) + 4.598 \times 10^{-5}$
	$\dot{x}_1$	$2.654411 \log(x_0) - 0.239982 \cos(x_1) - 0.112718(x_0 x_1) + 1.348938$
DeepSeek-V3.2	$\dot{x}_0$	$-6.490 \times 10^{-6}x_0 - 1.000029 \sin(x_1) - 0.199997x_0^2 - 1.215 \times 10^{-5}(x_0 \cos(x_1)) + 7.643 \times 10^{-5} \sin(2x_1) + 1.517 \times 10^{-5}$
	$\dot{x}_1$	$-1.000027 \frac{\cos(x_1)}{x_0} + 0.999992x_0 + 3.221 \times 10^{-4} \frac{\cos(2x_1)}{x_0} + 5.269 \times 10^{-5}$
Grok 4.1 Fast	$\dot{x}_0$	$-1.000030 \sin(x_1) - 0.199998x_0^2 + 7.786 \times 10^{-5} \sin(2x_1) - 4.216 \times 10^{-6}(x_0^2 \cos(x_1)) + 5.934 \times 10^{-6}$
	$\dot{x}_1$	$-1.000170 \frac{\cos(x_1)}{x_0} + 0.999566x_0 + 4.069 \times 10^{-4} \frac{\cos(2x_1)}{x_0} + 2.158 \times 10^{-4}(x_0 \log(x_0)) + 6.012 \times 10^{-4}$

G. Robustness analyses

G.1. Sigma-noise robustness

This section reports a sigma-noise robustness comparison on the 2D dimensionless Glider system (ID 2). DoLQ stores one system-level run per sigma, whereas ICSR, LASR, and LLM-SR store per-dimension outputs. To ensure a fair comparison, all final discovered equations are re-evaluated under a common protocol. Following the paper definition, the ID regime corresponds to the trajectory over $t_{\text{start}} \sim t_{\text{end}}$, while the ID-Ext regime corresponds to the full trajectory over $t_{\text{start}} \sim t_{\text{ood_end}}$. For each noise level σ and state dimension, NMSE is averaged across initial conditions. Tables 19, 20, and 21 jointly summarize the averaged metrics, per-dimension metrics, and final discovered equations under these noise levels.

The results indicate that DoLQ does not intrinsically filter observation noise; its absolute accuracy deteriorates as σ increases, despite remaining competitive with baseline methods. Robust equation discovery under noisy trajectories therefore remains

a clear limitation and an important direction for future work.

Table 19. Average NMSE measured across different sigma-noise levels for the dimensionless Glider system (ID 2). DoLQ stores system-level runs, whereas ICSR, LASR, EDL, and LLM-SR store method-specific outputs; all values are re-evaluated under the same ID and ID-Ext regimes. Cases where optimization exceeded the 5-minute time limit or the loss became ∞ are reported as NaN.

Method	$\sigma = 0.001$		$\sigma = 0.01$		$\sigma = 0.1$	
	ID	ID-Ext	ID	ID-Ext	ID	ID-Ext
DoLQ	0.1086	0.1246	0.6135	0.7668	32.5799	50.3185
ICSR	0.1289	0.2188	1.1742	2.0300	1.2066	2.1584
LASR	6.0070	6.0182	0.5390	0.6768	NaN	NaN
EDL	NaN	NaN	NaN	NaN	NaN	NaN
LLM-SR	4.6368	4.8001	370.2495	1.04e+04	2.17e+04	1.40e+05

Table 20. Per-dimension NMSE measured across different sigma-noise levels for the dimensionless Glider system (ID 2). All values are recomputed from the final discovered equations using the same ID and ID-Ext evaluation protocol. Cases where optimization exceeded the 5-minute time limit or the loss became ∞ are reported as NaN.

Method	σ	Dim	ID	ID-Ext
DoLQ	0.001	$\dot{x}_0$	0.0052	0.0078
		$\dot{x}_1$	0.2120	0.2413
	0.01	$\dot{x}_0$	0.4654	0.6868
		$\dot{x}_1$	0.7616	0.8469
	0.1	$\dot{x}_0$	12.5009	14.2665
		$\dot{x}_1$	52.6590	86.3704
ICSR	0.001	$\dot{x}_0$	0.0048	0.0100
		$\dot{x}_1$	0.2530	0.4276
	0.01	$\dot{x}_0$	0.5786	0.5851
		$\dot{x}_1$	1.7697	3.4750
	0.1	$\dot{x}_0$	0.5829	0.5762
		$\dot{x}_1$	1.8304	3.7405
LASR	0.001	$\dot{x}_0$	11.8347	11.8571
		$\dot{x}_1$	0.1792	0.1792
	0.01	$\dot{x}_0$	0.6409	0.6412
		$\dot{x}_1$	0.4371	0.7125
	0.1	$\dot{x}_0$	NaN	NaN
		$\dot{x}_1$	0.2028	0.2428
EDL	0.001	$\dot{x}_0$	NaN	NaN
		$\dot{x}_1$	NaN	NaN
	0.01	$\dot{x}_0$	NaN	NaN
		$\dot{x}_1$	NaN	NaN
	0.1	$\dot{x}_0$	NaN	NaN
		$\dot{x}_1$	NaN	NaN
LLM-SR	0.001	$\dot{x}_0$	5.5721	5.8846
		$\dot{x}_1$	3.7015	3.7157
	0.01	$\dot{x}_0$	511.0606	1.89e+04
		$\dot{x}_1$	229.4384	2035.1681
	0.1	$\dot{x}_0$	4.31e+04	2.79e+05
		$\dot{x}_1$	373.0867	1072.5261

Different LLM backbones can converge to slightly different symbolic forms due to disparities in mathematical inductive bias and because early term proposals continually alter subsequent search trajectories in iterative mechanisms like DoLQ. Several distinct LLM backbones—Gemini 2.5 Flash-Lite, GPT-4o mini, DeepSeek-V3.2, and Grok 4.1 Fast—were evaluated to assess DoLQ’s robustness across providers. Empirically, Gemini 2.5 Flash-Lite, DeepSeek-V3.2, and Grok 4.1 Fast successfully recovered the Glider system dynamics with only minor discrepancies in extraneous terms or coefficients. Conversely, GPT-4o mini frequently yielded qualitatively mismatched structures and inferior integral NMSE. For established

benchmarks, discovery performance is evaluated primarily by NMSE and secondarily by term-level agreement with the ground truth; for entirely unknown systems, deeper scientific validation would be essential to confirm the physical validity of the inferred structures.

G.2. Robustness to noisy descriptions

To assess sensitivity to description noise, DoLQ was evaluated on the Glider(2D) system with modified natural-language descriptions but fixed numerical trajectories. When provided with a partially mismatched description, DoLQ retained certain meaningful components, such as $\sin(x_1)$ and $1/x_0$, yet failed to completely recover the ground-truth equations; average residual NMSE degraded to 1.19×10^{-2} (ID) and 7.37×10^{-2} (ID-Ext). Under a fully mismatched description, the core dynamics of x_1 ($x_0 - \cos(x_1)/x_0$) were completely lost, and the residual NMSE further increased to 3.15×10^{-2} (ID) and 1.83×10^{-1} (ID-Ext). This suggests that DoLQ inherently trusts the provided qualitative description and lacks an intrinsic mechanism to correct erroneous prior information.

H. Reasoning-trace analysis

H.1. Hallucination audit of reasoning traces

We audited one representative Glider(2D) run at the log level and found that the most common hallucination pattern is premature mechanistic storytelling. Representative exploratory terms included $x_0 \cos(x_1)$, $\log(x_0)$, and $\tanh(x_0)$. These terms were sometimes assigned confident physical narratives before sufficient numerical evidence had accumulated. This effect appeared most clearly when the prompts required explicit semantic justification for exploratory terms whose physical meaning was only weakly grounded by the system description. In our audit, these narratives affected search direction but not final acceptance, because every candidate still had to pass coefficient optimization, residual evaluation, and iterative comparison against competing candidates. The over-interpreted terms were not retained in the final discovered equation.

I. Time complexity analysis

The computational cost of DoLQ is analyzed by decoupling LLM token processing from numerical parameter optimization. As these operations utilize fundamentally different resources, merging them into a single asymptotic bound would obfuscate the system’s true computational load. Table 22 outlines the notation adopted for this complexity analysis.

LLM cost. At each DoLQ cycle, the Sampler Agent and Scientist Agent are each invoked once. If their total input and output token counts are denoted by L_{samp} and L_{sci} , respectively, then the token-processing cost per cycle is

$$O(L_{\text{samp}} + L_{\text{sci}}). \quad (3)$$

The Scientist Agent can reduce practical runtime by filtering poor structures early, but this affects the average search trajectory rather than the worst-case asymptotic form of a single LLM call.

Parameter optimization cost. For the j -th dimension, one evaluation of the residual objective in Eq. (2) over N time points costs $O(Nk_j)$, assuming each symbolic primitive in a candidate term is evaluated in constant time. Differential evolution therefore incurs

$$O(G \cdot NP \cdot N \cdot k_j). \quad (4)$$

Using dense BFGS refinement adds repeated residual evaluations together with quasi-Newton updates over p_j trainable parameters, yielding

$$O(I_j(Nk_j + p_j^2)). \quad (5)$$

Since DoLQ evaluates BFGS alone, differential evolution alone, and the hybrid differential-evolution-plus-BFGS strategy before selecting the best result, the constant factor increases, but the asymptotic optimization cost per dimension remains

$$O((G \cdot NP + I_j)Nk_j + I_j p_j^2). \quad (6)$$

Scientist quantitative evaluation cost. The quantitative evaluation in Algorithm B.3 performs one baseline residual evaluation and then one additional residual evaluation per term after temporarily zeroing its coefficient. For the j -th

dimension, this requires $k_j + 1$ residual evaluations, so the total cost is

$$O((k_j + 1)Nk_j) = O(Nk_j^2). \quad (7)$$

This term is separate from the LLM reasoning cost and must be counted explicitly in the overall complexity.

Lightweight bookkeeping. Function construction, JSON parsing, and keep/hold/remove bookkeeping are linear in the number of proposed terms, i.e.,

$$O\left(\sum_{j=1}^{d_{\text{sys}}} k_j\right), \quad (8)$$

which is dominated by parameter optimization and ablation.

Overall complexity. For one DoLQ cycle that evaluates H candidate hypotheses, the numerical computation scales as

$$O\left(H \sum_{j=1}^{d_{\text{sys}}} [(G \cdot NP + I_j)Nk_j + I_j p_j^2 + Nk_j^2]\right), \quad (9)$$

while the LLM token-processing cost scales as

$$O(L_{\text{samp}} + L_{\text{sci}}). \quad (10)$$

Over T search cycles, the total complexity becomes

$$O(T(L_{\text{samp}} + L_{\text{sci}})) \quad (11)$$

for token processing and

$$O\left(TH \sum_{j=1}^{d_{\text{sys}}} [(G \cdot NP + I_j)Nk_j + I_j p_j^2 + Nk_j^2]\right) \quad (12)$$

for numerical computation.

Practical implication. The main numerical advantage of DoLQ comes from using residual MSE during optimization. Unlike integral-based objectives such as Eq. (1), Eq. (2) does not require solving the ODE at every optimizer step, so the inner loop is reduced to direct residual evaluation on sampled data. Integral NMSE remains useful for final evaluation, but it is not part of the optimization loop analyzed above. Similarly, early filtering by the Scientist Agent improves practical efficiency by reducing the average number of poor candidates and unnecessary terms, rather than by changing the worst-case asymptotic order.

J. Efficiency analysis

To ensure an equitable baseline comparison, LLM-SR’s runtime metrics are aggregated across all target dimensions, reflecting its dimension-wise execution paradigm compared to DoLQ’s joint system-level discovery. Evaluations were performed in identical environments (Intel Core i7-12700K, 20 logical cores, 94 GiB RAM). Table 23 details the empirical CPU-time, wall-clock duration, total elapsed time, and peak memory usage.

Given LLM-SR’s decoupled execution, direct comparison necessitates aggregating its CPU and temporal metrics while extracting the highest peak memory recorded across per-dimension runs. The maximum value accurately captures the instantaneous memory ceiling rather than an artificial sum. Consequently, while DoLQ does not uniformly outperform baselines in raw CPU computation time, it exhibits systematically lower wall-clock duration, total elapsed time, and peak memory footprints.

Floating-point operations (FLOPs) are intentionally omitted because the hybrid pipeline integrates remote LLM inference with local numerical optimization, rendering hardware-agnostic FLOP estimation both ill-defined and challenging to reproduce. Efficiency is thus benchmarked primarily through computation time and memory allocation.

K. Prompts and responses

K.1. Sampler agent prompt and response

We present example prompts and outputs for the Sampler and Scientist agents in Figure 12 through Figure 15. The labeled sections in these figures explicitly correspond to the components illustrated in the framework overview (Figure 2), detailing the specific instructions and constraints provided to each agent during the discovery process.

K.2. Scientist agent prompt and response

Figure 14 and Figure 15 present a representative Scientist-agent prompt/response pair, showing how accumulated insights, experiment summaries, and evaluation instructions are organized before the model returns term-level judgments and actions.

K.3. Initial sampler prompt at iteration 1

Figure 16 shows the initialization-time Sampler prompt used before any prior evaluations are available, making explicit what context is given when accumulated knowledge, term-level feedback, and removed-term history are still empty.

Initial Sampler Prompt (Iteration 1)

You are a helpful assistant tasked with discovering mathematical term structures for scientific systems. Complete the term list below, considering physical meaning and relationships of inputs.

System Description
 The glider is viewed as an idealized system whose motion is expected to arise from the interplay of gravity and aerodynamic forces, with speed and flight path angle evolving from assumed initial conditions.

SCIENTIST AGENT GUIDANCE
 The Scientist agent has analyzed previous experiments and provides the following guidance:

Accumulated Knowledge (Theory)
 None. This is the first iteration, so no accumulated insight had been produced yet.

Term-by-Term Evaluation (Previous Attempt Analysis)
 Evaluation results for each term. keep = retain, hold = hold/modify, remove = eliminate:
 None. There was no previous attempt because this prompt was given before any sampler output had been evaluated.

Removed Terms List (Ban List)
 The following term structures have negatively impacted performance. **Do NOT propose them again:**
 None. At iteration 1, no term skeleton had yet been removed.

Goal: Reflect the Scientist's insights and guidance in the equation structure.
 [Required Conditions (Violation Will Cause Errors)]

1. You can use: `import numpy as np`
2. Target System Context: Input variables are `x0`, `x1`.
 -- This system is 2-dimensional
 -- Variables `x2` and above do not exist.
3. Term Format: Propose terms WITHOUT coefficients. The system will automatically attach trainable parameters.
 -- Correct: `"x0"`, `"np.sin(x0)"`, `"x0*x1"`
 -- Incorrect: `"params[0]*x0"`, `"C*x0"`, `"0.5*x0"`
4. Term Complexity: You MAY use internal constants if they have physical meaning (e.g., frequency, phase).
 -- Example: `"np.sin(2*x0)"` is allowed and encouraged if the factor 2 is significant.
 -- Note: The system will still attach an outer trainable parameter (e.g., `params[0]*np.sin(2*x0)`).
5. Symbolic Constants: Do NOT use symbolic constants like `'g'`, `'k'`, `'m'`. Use numerical values.
 -- Correct: `"9.81*x0"` (if `g=9.81` is known), `"np.pi*x0"`
 -- Incorrect: `"g*x0"` (will cause `NameError`)
6. No duplicates: Equations identical to previous attempts are forbidden. Structural modifications are required.
7. Reasoning required: When proposing each term, provide a physical/mathematical reasoning based on the system description (desc).

[Example (2D System)]
`x0.t: ["x0", "x1"]`
`x1.t: ["x0", "np.sin(x1)"]`

Figure 16. Initial prompt to the Sampler agent at iteration 1, where accumulated knowledge, term-level evaluation, and removed-term history are empty at initialization.

Sampler Prompt

You are a helpful assistant tasked with discovering mathematical term structures for scientific systems. Complete the 'term.list' below, considering the physical meaning and relationships of inputs.

System Description

Glider flight experiment. State variables: forward velocity x_0 , path angle x_1 , horizontal position x_2 , vertical altitude x_3 . [... omitted...]

SCIENTIST AGENT GUIDANCE

The Scientist agent has analyzed previous experiments and provides the following guidance:

Accumulated Knowledge (Theory)

- The kinematic terms for horizontal and vertical position ($x_2.t$ and $x_3.t$) show strong physical relevance, particularly components involving x_0 and trigonometric functions of x_1 . The damping term in $x_0.t$ and the velocity-dependent term in $x_1.t$ demonstrate good physical grounding. However, constant terms lack clear physical justification.

(1a)

Term-by-Term Evaluation (Previous Attempt Analysis)

Evaluation results for each term. keep = retain, hold = hold/modify, remove = eliminate:

$x_0.t$:

- $C*(x_0)$: KEEP
- $C*(np.\sin(x_1))$: KEEP
- $C*x_0**3$: REMOVE

$x_1.t$:

- $C*(np.\cos(x_1))$: KEEP
- $C*(x_0)$: HOLD
- $C*x_1**3$: REMOVE

$x_2.t$:

- $C*(x_0*np.\cos(x_1))$: KEEP

$x_3.t$:

- $C*(x_0*np.\sin(x_1))$: KEEP
- $C*x_0*x_1*x_2$: REMOVE

Removed Terms List (Ban List)

The following term structures have negatively impacted performance. Do NOT propose them again:

$x_0.t$: $C*(9.81)$, $C*(np.\cos(x_1))$, $C*(x_0*x_1)$

$x_1.t$: $C*(1)$, $C*(np.\sin(x_1))$, $C*(x_0*x_1)$, $C*(x_3)$

$x_2.t$: $C*(9.81)$, $C*(x_1)$

$x_3.t$: $C*(x_0)$, $C*(x_1)$, $C*(x_3)$

(1b)

Goal: Reflect the Scientist's insights and guidance in the equation structure.

[Required Conditions (Violation Will Cause Errors)]

1. You can use: import numpy as np
2. Target System Context: Input variables are x_0 , x_1 , x_2 , x_3 .
 - This system is 4-dimensional
 - Variables x_4 and above do not exist.
3. Term Format: Propose terms WITHOUT coefficients. The system will automatically attach trainable parameters.
 - Correct: " x_0 ", " $np.\sin(x_0)$ ", " x_0*x_1 "
 - Incorrect: " $params[0]*x_0$ ", " $C*x_0$ ", " $0.5*x_0$ "
4. Term Complexity: You MAY use internal constants if they have physical meaning (e.g., frequency, phase).
 - Example: " $np.\sin(2*x_0)$ " is allowed and encouraged if the factor 2 is significant.
 - Note: The system will still attach an outer trainable parameter (e.g., $params[0]*np.\sin(2*x_0)$).
5. Symbolic Constants: Do NOT use symbolic constants like 'g', 'k', 'm'. Use numerical values.
 - Correct: " $9.81*x_0$ " (if $g=9.81$ is known), " $np.pi*x_0$ "
 - Incorrect: " $g*x_0$ " (will cause NameError)
6. No duplicates: Equations identical to previous attempts are forbidden. Structural modifications are required.
7. Reasoning required: When proposing each term, provide a physical/mathematical reasoning based on the system description (desc).

[Example (4D System)]

$x_0.t$: [$"x_0"$, $"x_1*x_2"$, $"x_3"$]

$x_1.t$: [$"x_0"$, $"np.\sin(x_1)"$]

$x_2.t$: [$"x_0*x_1"$]

$x_3.t$: [$"-9.81"$, $"x_0"$]

(1c)

Figure 12. The Sampler Prompt. The labeled sections (1a)-(1c) correspond to the Sampler components in Figure 2: (1a) System Description & Insight, (1b) Feedback & Constraints, and (1c) Action & Instruction.

Sampler LLM Response

```
{
  "ode_pairs": [
    {
      "x0.t": ["x0", "np.sin(x1)", "x0*x0"],
      "x1.t": ["np.cos(x1)", "x0", "x0*x0"],
      "x2.t": ["x0*np.cos(x1)", "x0"],
      "x3.t": ["x0*np.sin(x1)", "-9.81", "x2*x2"]
    },
    [... other pairs omitted...]
  ]
}
```

Figure 13. Example response from the Sampler LLM showing the structured JSON response containing candidate ODE terms for each dimension.

Scientist Prompt

You are a senior scientist specializing in ODE discovery. Your role is to evaluate proposed mathematical terms and provide guidance to improve the term.list in the next iteration.

Progress: Currently on iteration 50 of 100 total

System Description:

Glider flight experiment. State variables: forward velocity x_0 , path angle x_1 , horizontal position x_2 , vertical altitude x_3 . [... omitted...]

(2a)

Accumulated Insights:

The kinematic terms for $x_{2.t}$ and $x_{3.t}$ show strong physical relevance, particularly components involving x_0 and trigonometric functions of x_1 . The damping term in $x_{0.t}$ and the velocity-dependent term in $x_{1.t}$ demonstrate good physical grounding. [... omitted...]

(2b)

Previous Experiment Results:

[Global Best] ($x_{0.t}$: $2.187e-02$, $x_{1.t}$: $6.520e-01$, $x_{2.t}$: $2.846e-07$, $x_{3.t}$: $2.005e-07$)
 $x_{0.t} = -0.2688*(x_0) - 9.8336*(np.sin(x_1)) - 0.0279*(x_0*np.cos(x_1)) + 0.4777$
 [... other dimensions omitted...]

Current Attempt: ($x_{0.t}$: $2.761e-02$, $x_{1.t}$: $3.555e+00$, $x_{2.t}$: $2.846e-07$, $x_{3.t}$: $2.086e-07$)
 $x_{0.t} = -0.2982*(x_0) - 9.8333*(np.sin(x_1))$
 [... other dimensions omitted...]

(2c)

Task: Perform term-by-term evaluation.

For all terms in the current attempt, evaluate the following 2 aspects:

1. Semantic Quality:

- **good:** Clearly aligns with the physical/mathematical meaning (**Max 3 per function**)
- **neutral:** Has some relevance but not essential
- **bad:** Unrelated to or contradicts the system

2. **Reasoning:** Explain your evaluation in 1-2 sentences. Focus on physical meaning.

Important Notes:

- Use coefficient analysis: Terms with coefficients near 0 are removal candidates.

(2d)

Figure 14. The Scientist Prompt. The labeled sections (2a)-(2d) correspond to the Scientist components in Figure 2: (2a) Role & Description, (2b) Accumulated Insights, (2c) Experiment Results, and (2d) Evaluation Task.

Scientist LLM Response

```
{
  "x0_t": [
    {"term": "C*(x0)", "semantic_quality": "good", "action": "keep"},
    {"term": "C*(np.sin(x1))", "semantic_quality": "good", "action": "keep"},
    {"term": "C*(x0*x0)", "semantic_quality": "neutral", "action": "hold"}
  ],
  [... x1_t, x2_t, x3_t omitted...]
}
```

Figure 15. Example response from the Scientist LLM showing term-by-term evaluation with semantic quality assessment and action recommendations.

Table 21. Final discovered equations under different noise levels for the dimensionless Glider system (ID 2). A dash (-) indicates that no valid equation was available for reporting.

Method	σ	Dim	Equation
DoLQ	0.001	$\dot{x}_0$	$((((-0.17845980618743792 \cdot (x_0^2)) + (-1.0941877346694517 \cdot \sin(x_1))) + (0.05314606941174399 \cdot x_1)) + (-0.3391220551372564))$
		$\dot{x}_1$	$((((-1.8940271186983297 \cdot (-9.81)) + (-0.8283632477740318 \cdot \cos(x_1))) + (-0.20272251166224334 \cdot (x_1/x_0))) + (-15.836236841336438))$
	0.01	$\dot{x}_0$	$((((-10.284937863923174 \cdot 9.81) + (0.4911498533070019 \cdot x_1)) + (-1.9129771153833335 \cdot \sin(x_1))) + 97.78128237249153)$
		$\dot{x}_1$	$((((-1.3512098187497759 \cdot \sin(x_1)) + (2.080726220812281 \cdot x_0)) + (-0.16870678464779224 \cdot (x_1 \sin(x_0)))) + (-2.1465100178836565))$
	0.1	$\dot{x}_0$	$(((((4.610553824850981 \cdot \sin(x_1)) + (-12.839898886682368 \cdot (x_0 \sin(x_1)))) + (2.454264804272983 \cdot (x_0^2 \sin(x_1)))) + 1.9735115401721728)$
		$\dot{x}_1$	$((((-0.8346838384737242 \cdot (x_1 \cos(x_1))) + (-4.785671048569711 \cdot (x_0 x_1))) + (1.7968322398349674 \cdot (x_0^2 x_1))) + 9.272749877208437)$
ICSR	0.001	$\dot{x}_0$	$-0.185x_0^2 + 0.0035x_0x_1 + 0.0041x_1^2 - 1.0961 \sin(x_1) - 0.1851$
		$\dot{x}_1$	$0.5688x_0 - 0.0262x_1^2 + 2.0966 - 1.0063/x_0$
	0.01	$\dot{x}_0$	$-0.9509x_0 \sin(0.9597x_1) + 0.0595x_1 - 0.822$
		$\dot{x}_1$	$-1.2089x_1 \cos(0.4621x_0) + 5.343$
	0.1	$\dot{x}_0$	$-0.9509x_0 \sin(0.9597x_1) + 0.0595x_1 - 0.822$
		$\dot{x}_1$	$-1.2089x_1 \cos(0.4621x_0) + 5.343$
LASR	0.001	$\dot{x}_0$	$-4.162585368621153/x_1$
		$\dot{x}_1$	$\log(\sinh(x_0))$
	0.01	$\dot{x}_0$	$-\sin(x_1) - 0.4299814235724901$
		$\dot{x}_1$	$\log(x_0 \cdot \cosh(x_0) - 1.5820667442977299) + 0.2913439363183297$
	0.1	$\dot{x}_0$	$-1.339672350584124 \cdot \tan(\exp(x_0/x_1)/12.367644780162692)$
		$\dot{x}_1$	x_0
EDL	0.001	$\dot{x}_0$	-
		$\dot{x}_1$	-
	0.01	$\dot{x}_0$	-
		$\dot{x}_1$	-
	0.1	$\dot{x}_0$	-
		$\dot{x}_1$	-
LLM-SR	0.001	$\dot{x}_0$	$5.7922x_1 + 0.3353x_0 - 1.2690x_0^2 - 10.4843 - 0.9018x_1^2 + 0.2287x_0^3 + 0.0289x_0x_1^2 + 0.0321x_0^3x_1 + 0.0383x_1^3 - 0.0019x_0^2x_1^3$
		$\dot{x}_1$	$-1.6745x_0 - 0.6770 \cos(x_1)/(x_0 + 10^{-9}) - 0.2463x_0 \sin(x_1) + 1.9791x_0^2 - 0.2591x_0^3 + 0.0039x_1^2 + 0.3814x_0x_1 - 0.2232x_0^2x_1 - 0.2732/(x_0 + 10^{-9}) - 0.2510 \sin(2x_1)$
	0.01	$\dot{x}_0$	$-4.1139x_0^2 + 0.1036x_1^2 + 2.3645x_0x_1 + 0.0819x_0^2x_1 - 0.1884x_0x_1^2 + 0.7252x_0^3 - 0.0395x_1^3 - 1.9019 \sin(\text{clip}(x_0)) - 1.5588 \log(\text{clip}(x_0)) + 0.0039 \exp(x_1)$
		$\dot{x}_1$	$[-1.3646x_0^2 \cos(x_1) \sin(x_1) - 3.1803x_0^2 \sin(x_1) + 12.0188 \cos(x_1) + 57.1401 - 3.1391x_0 \sin(x_1) + 1.0047x_0 \cos(x_1) + 0.2624x_1^3]/(x_0 + 10^{-6}) + 57.1401/(x_0 + 10^{-6}) - 30.5360x_1/(x_0 + 10^{-6}) + 0.5511x_0x_1 \sin(x_1)$
	0.1	$\dot{x}_0$	$\frac{0.5 \cdot 0.813984 \cdot x_0^2 \cdot 0.813984 \cdot \max(-125.0561 + 0.0961(-44.7349 + 22.1524x_1 - 0.2354x_1^3)^2, 0.01)}{0.769795} \cos(x_1) - \frac{0.5 \cdot 0.813984 \cdot x_0^2 \cdot 0.813984 \cdot 0.004265(-44.7349 + 22.1524x_1 - 0.2354x_1^3) \sin(x_1)}{0.769795} - 3.9405 \sin(x_1)$
		$\dot{x}_1$	$-10.8801x_1 + 0.1683x_1^3 - 9.8663 \sin(x_1) - 44.2731x_0 - 0.1425x_0^2 - 87.9659/x_0 + 11.9680/x_0^2 - 15.7989x_0x_1 + 5.6508x_0^2x_1 + 199.1791$

Table 22. Notation used in the time complexity analysis of DoLQ.

Symbol	Meaning
T	Total number of DoLQ search cycles
H	Number of candidate hypotheses evaluated per cycle
d_{sys}	Number of state dimensions in the ODE system
N	Number of sampled time points used in the residual objective
k_j	Number of symbolic terms in the j -th equation
p_j	Number of trainable parameters in the j -th equation
G	Number of differential evolution generations
NP	Differential evolution population size
I_j	Number of BFGS refinement iterations for dimension j
L_{samp}	Total input and output tokens processed by the Sampler Agent per cycle
L_{sci}	Total input and output tokens processed by the Scientist Agent per cycle

Table 23. CPU, runtime, and peak-memory comparison of DoLQ and LLM-SR. Total CPU time is the summed CPU time, total wall time is the wall-clock runtime, end-to-end elapsed time is the full elapsed time, and peak memory is the maximum resident memory usage in GiB.

Problem (ID)	Method	Total CPU time (s)	Total wall time (s)	End-to-end elapsed (s)	Peak memory (GiB)
2	DoLQ	12.169	568.095	608	0.234
	LLM-SR	6.781	3080.691	3080.691	0.421
3	DoLQ	12.300	744.724	775	0.235
	LLM-SR	6.848	2293.902	2296.985	0.420
8	DoLQ	12.660	712.452	746	0.238
	LLM-SR	14.363	6406.439	6411.136	0.422